

CORE MATHEMATICS GRADE 10: SURFACE AREA AND VOLUME OF SOLIDS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.7 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.7: Surface Area and Volume of Solids
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Surface area of prisms (triangular, rectangular, and other prisms) – Surface area of pyramids (square-based and other pyramids) – Surface area of cones – Surface area of frustums (cone frustums) – Surface area of spheres – Surface area of composite solids – Volume of prisms (triangular, rectangular, and other prisms) – Volume of pyramids (square-based and other pyramids) – Volume of cones – Volume of frustums – Volume of spheres – Volume of composite solids – Real-life applications of surface area and volume of solids
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) determine the surface area of prisms, pyramids, cones, frustums and spheres, b) determine the surface area of composite solids, c) determine the volume of prisms, pyramids, cones, frustums and spheres, d) determine the volume of composite solids, e) apply surface area and volume of solids to real-life situations, f) appreciate the use of surface area and volume of solids in real-life situations.
Core Competencies	– Communication and Collaboration: the learner discusses and shares findings with peers when measuring and calculating the surface area and volume of locally available solid models such as tins, packets and containers used for ugali flour or cooking oil. – Critical Thinking and Problem Solving: the learner analyses composite solid shapes found in real-life Kenyan contexts — such as grain silos in Eldoret or water tanks in rural Rift Valley — and decomposes them to determine total surface area and volume. – Creativity and Imagination: the learner improvises and constructs nets and models of prisms, pyramids, cones and spheres

	using locally available materials (cardboard, manila paper) to verify surface area formulae. – Digital Literacy: the learner uses digital devices and software (where available) to explore 3D solid models, nets and volume visualisations. – Self-Efficacy: the learner builds confidence in applying surface area and volume formulae independently to solve real-world packaging, construction and storage problems relevant to Kenyan industries.
Core Values	– Responsibility: the learner carefully handles models of solids and measurement tools collected from the immediate environment, and accurately records measurements to ensure correct calculations of surface area and volume. – Integrity: the learner applies formulae with accurate measurements and honest working, adhering to mathematical rigour when solving surface area and volume problems. – Unity: the learner works collaboratively in groups to construct solid models, measure dimensions and verify calculated surface areas and volumes, appreciating each member's contribution. – Respect: the learner recognises and values the input of every group member during practical activities involving the construction and measurement of solid models.
Science & Engineering Practices	– Using Mathematics and Computational Thinking: learners apply and manipulate formulae for surface area and volume of prisms, pyramids, cones, frustums, spheres and composite solids, using arithmetic and algebraic reasoning to solve quantitative problems. – Planning and Carrying Out Investigations: learners plan and conduct hands-on measurement investigations using real solid objects (tins, boxes, cones) to collect dimension data, then compute and compare calculated surface area and volume against estimates. – Analysing and Interpreting Data: learners examine measured dimensions and computed results, identify errors or inconsistencies, and draw conclusions about which solid shape optimises volume for a given surface area — relevant to packaging and storage design. – Constructing Explanations: learners construct clear mathematical explanations of how surface area and volume formulae are derived from nets and cross-sectional areas, and explain results in context (e.g. why a cylindrical water tank in Kisumu holds more than a box of the same surface area). – Obtaining, Evaluating, and Communicating Information: learners present their findings on surface area and volume of composite solids to the class, evaluating the reasonableness of results and communicating solution strategies clearly.
Pertinent & Contemporary Issues (PCIs)	– Education for Sustainable Development (Environmental Education): the learner uses and repurposes locally available materials such as cardboard packaging, tin cans and plastic containers to construct solid models, promoting environmental consciousness and waste reduction. – Social Cohesion: the learner works in groups to build models, measure solids and solve problems, fostering teamwork and mutual respect across diverse learner backgrounds. – Socio-Economic Issues: the learner applies surface area and volume concepts to real Kenyan economic contexts such as determining the amount of metal sheet needed to fabricate a water tank in Nakuru, or the capacity of a grain store in the Rift Valley, linking mathematics to livelihoods and national development. – Home Science and Technology: the learner connects volume calculations to everyday household decisions such as estimating the volume of a sufuria used for cooking ugali or the capacity of a storage jerrycan.
Career Connections	– Architecture and Civil Engineering: understanding surface area informs material estimation for roofing, wall cladding and flooring of buildings — directly relevant to Kenya's affordable housing programmes. – Manufacturing and Packaging Industry: volume and surface area calculations are essential for designing and optimising packaging for Kenyan food products such as maize flour, cooking oil and tea from Kericho. – Water and Sanitation Engineering: volume of cylinders, frustums and spheres is applied in designing water storage tanks,

	pipes and reservoirs for communities around Lake Victoria and arid regions. – Agriculture and Agribusiness: farmers in the Rift Valley and Central Kenya use volume concepts to estimate grain storage capacity in cylindrical silos and conical heaps. – Medicine and Pharmacy: volume of spherical capsules and cylindrical vials is critical in drug dosage calculations and medical device manufacturing. – Surveying and Land Management: surface area concepts underpin calculations of land volumes in road construction and mining operations in regions such as Taita Hills.
Focus for Lessons	This unit develops learners' ability to calculate the surface area and volume of common solids — prisms, pyramids, cones, frustums, spheres and composite solids — grounding every formula in the physical reality of objects found in Kenyan daily life (grain tins, water tanks, roofing cones, sports balls). Learners move from hands-on net construction and measurement of real objects to formal derivation and application of formulae, building both procedural fluency and conceptual understanding. The unit culminates in a real-life project where learners design or evaluate a practical solid structure (e.g. a water storage tank or grain silo) relevant to their local community, integrating surface area and volume to make informed decisions.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does the same amount of metal sheet make containers that hold very different amounts of grain — and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction? KICD KEY INQUIRY QUESTIONS: 1. How do we determine the surface area and volume of solids? 2. How are surface area and volume of solids applied in real-life situations?
Anchoring Phenomenon	A local posho mill owner in Eldoret has two large metal storage containers for holding maize grain: one is a tall cylindrical drum and the other is a rectangular metal box. Both containers appear roughly the same size from the outside, and the metalsmith who made them used the same total area of iron sheet for each. Yet when grain is poured in, the cylindrical drum holds noticeably more maize than the rectangular box. The mill owner is puzzled — if the same amount of metal was used for both, why does one hold more grain? Students observe photographs and physical models of both containers, measure their dimensions, and are challenged to explain the discrepancy. This phenomenon is puzzling because learners intuitively assume equal surface area means equal volume, but discover that shape — not just material used — determines storage capacity, launching an investigation into how surface area and volume are related yet distinct properties of three-dimensional solids.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Surface Area of Prisms: Learners observe the anchoring phenomenon (two containers, same metal sheet, different grain capacity). They predict which holds more and why. They construct nets of rectangular and triangular prisms from cardboard, measure faces, and derive the surface area formula. They calculate the surface area of the rectangular box from the phenomenon and record findings on the Driving Question Board (DQB). Lesson 2 — Surface Area of Pyramids: Learners examine pyramid-roofed structures common in Kenyan homesteads (e.g. traditional Maasai manyatta roofs and school hall rooftops). They construct nets of square-based and triangular pyramids, identify slant height versus vertical height, and derive the surface area formula. They solve problems involving material needed to cover a pyramid-shaped roof. Lesson 3 — Surface Area of Cones and Frustums: Learners investigate the cone shape using real objects (ice-cream cones, conical grain heaps, funnel tops of water tanks). They derive the curved surface area formula using the net of a cone (a sector of a circle). They extend this to the frustum by considering a cone with its top removed, linking to the Turkana water-tank supporting phenomenon. Learners calculate the metal sheet needed for a conical tank top.

	Lesson 4 — Surface Area of Spheres and Composite Solids: Learners explore the surface area of spheres by wrapping a ball in paper strips, then verify using the formula $SA = 4\pi r^2$. They extend to composite solids (cylinder + cone; hemisphere + cylinder) and practise decomposing shapes. They connect to the Nairobi packaging factory supporting phenomenon and the football supporting phenomenon. The DQB is updated with new understanding. Lesson 5 — Volume of Prisms and Pyramids: Learners return to the anchoring phenomenon and now investigate volume. They fill rectangular and triangular prism models with sand or water and compare. They derive Volume = base area $\times$ height for prisms and $V = \frac{1}{3} \times \text{base area} \times \text{height}$ for pyramids by comparing filled models. They compute the volume of the rectangular box from Lesson 1 and compare it with the cylinder (previewing Lesson 6). Lesson 6 — Volume of Cones, Frustums and Spheres: Learners use the relationship between cone and cylinder volumes ($V_{\text{cone}} = \frac{1}{3} V_{\text{cylinder}}$) demonstrated by pouring water between matching models. They derive $V = \frac{1}{3}\pi r^2 h$ for cones and $V = \frac{4}{3}\pi r^3$ for spheres. They calculate the volume of the cylindrical drum from the anchoring phenomenon and formally resolve why it holds more grain than the rectangular box of equal surface area. Lesson 7 — Volume and Surface Area of Composite Solids & Real-Life Applications: Learners tackle multi-step problems involving composite solids such as a cylindrical water tank with a hemispherical top (common in Kenyan water projects). They calculate both the total surface area (painting/material cost) and volume (water storage capacity). They also work on a grain silo problem set in the Rift Valley, integrating all formulae learned. Lesson 8 — Project Presentations, Model Building & Unit Consolidation: Learner groups present their home/extended project: designing a storage container (grain silo, water tank or cooking pot) for a Kenyan community need, specifying dimensions, total surface area of material required and volume/capacity. Groups critique each other's designs. The class builds a final cumulative model on the DQB linking surface area and volume formulae to the anchoring phenomenon, resolving the driving question fully and discussing careers that use these concepts.
Supporting Phenomena	– A mandazi vendor in Mombasa notices that ball-shaped (spherical) mandazi cool down more slowly than flat disc-shaped ones of the same mass — prompting investigation into how surface area relative to volume affects heat loss and connects to the surface area of spheres. – A water harvesting project in Turkana County must choose between a conical, cylindrical and hemispherical roof-catchment tank; all use the same amount of plastic sheeting — students investigate which shape stores the most water, exploring volume and surface area of cones, cylinders and spheres. – Workers at a Nairobi packaging factory notice that when a conical top is placed on a cylindrical can (like a composite solid), the total surface area to be painted changes in a non-obvious way — prompting learners to decompose composite solids and calculate total surface areas systematically. – A school in Kisumu receives a donation of spherical footballs and asks: how much leather (surface area) is needed to make each ball, and how much air (volume) does it hold? — anchoring the surface area and volume of spheres in a familiar, motivating context.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (40 minutes (one lesson; the rectangular-prism net and the 3-4-5 triangular prism net are done in pairs)): Anchoring Phenomenon & Surface Area of Prisms: Same Metal, Different Grain

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Launch the anchoring phenomenon (Mzee Kiplagat's two containers made from the same 30,800 cm ² of iron sheet but holding different amounts of maize) and build the first idea needed to explain it: the surface area of a prism is the sum of the areas of the faces in its net, which equals 2 x base area + perimeter of base x length.
Knowledge	By the end of the lesson the learner should be able to: (1) draw and cut out the net of a rectangular prism and of a triangular prism; (2) find the total surface area (TSA) of a prism by adding the areas of all faces in its net; (3) use $TSA = 2 \times \text{base area} + \text{perimeter of base} \times \text{length}$; (4) calculate the TSA of the rectangular box (50 cm x 50 cm x 129 cm) as 30,800 cm ² ; (5) state that surface area (material covering a solid) and volume (space inside) are different quantities. KICD outcome addressed: determine the surface area of prisms, pyramids, cones, frustums, spheres and composite solids. KICD Key Inquiry Question 1: How do we determine the surface area and volume of solids?
Skills	Developing and using models (nets and physical prism models); asking questions and defining problems (DQB); analysing and interpreting data (measuring faces, tabulating areas); using mathematics (calculating areas of rectangles and triangles); constructing explanations and arguing from evidence. KICD core competencies: Critical Thinking and Problem Solving (learners compare the two containers and test the idea that equal material means equal capacity); Communication and Collaboration (learners share measurements and reasoning in pairs and with the class); Creativity and Imagination (learners build nets from cardboard).
Attitudes	KICD values: Responsibility (learners take care of cardboard, scissors and measuring tools and return them); Integrity (learners record measurements honestly and accurately, as Wanjiru the metalsmith did by measuring and not guessing); Unity (learners work in pairs and respect each other's ideas). PCI: Education for Sustainable Development (learners reuse waste cardboard and cut with minimal wastage, like a metal fabricator who wastes no sheet).
Key Inquiry Question	1. How do we determine the surface area and volume of solids? 2. How are surface area and volume of solids applied in real-life situations? Lesson focus: If two containers use exactly the same amount of metal, must they hold the same amount of maize?
Purpose in Storyline	This is the opening lesson. It introduces the phenomenon and opens the DQB. It gives learners the sum-of-faces method (nets) for surface area and the box's TSA of 30,800 cm ² . Lesson 2 builds on nets and the sum-of-faces method for pyramids. Comparing volumes is left for Lessons 5 and 6, so learners leave this lesson knowing the box's SA but still puzzled about why the drum holds more.
Safety Notes	Scissors: cut away from the body, pass handles-first, and keep scissors on the desk when not cutting. Use blunt-tip scissors or a razor-free cutting method only. Do not use blades. Keep floors clear of cardboard offcuts to prevent slipping. Do not bring real metal drums to class; use photographs and small models. If using a real jerrican or tin, check for sharp edges first.

B. LESSON OVERVIEW

Learners meet Mzee Kiplagat, who owns a posho mill in Eldoret. He asked Wanjiru, a jua kali metalsmith, to make two closed storage containers for maize: a cylindrical drum and a square-based rectangular box, each from exactly 30,800 cm² of iron sheet. Both stand about 1 m tall and look about the same size, yet the drum holds more maize. Learners first predict which holds more and why, then investigate surface area by making nets of a 5 cm x 4 cm x 3 cm rectangular prism and a triangular prism with 3-4-5 cm right-angled triangle ends and length 10 cm. They add the face areas, discover that $TSA = 2 \times \text{base area} + \text{perimeter} \times \text{length}$, and apply it to the box (50 cm x 50 cm x 129 cm) to find 30,800 cm². They then record what they have learned and what they still wonder on the Driving Question Board (DQB). They leave with

the box's surface area confirmed but the puzzle of the drum still open, since the drum's SA and the volumes are investigated in later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Surface area using a net: triangular prism Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Surface Area of Triangular Prisms (Flexbook) Source: CK-12  Search ARES for similar readings	Learners view a photograph or poster of Mzee Kiplagat's posho mill in Eldoret showing the tall cylindrical drum and the square-based rectangular box side by side. The teacher reads the story aloud: Wanjiru, the jua kali metalsmith, used exactly 30,800 cm ² of iron sheet for each closed container, and both stand about 1 m tall. Individually (2 minutes) learners write on a prediction slip: (a) Which container will hold more maize, or will they hold the same? (b) Why do you think so? They draw a quick sketch of each container. They then share with a partner (2 minutes) and record a class tally of predictions on the board: drum holds more, box holds more, same. Learners are told only that the owner found that the two do not hold the same amount; the result is not revealed in detail yet.	Display the photograph and say: 'Mzee Kiplagat is puzzled. Wanjiru used exactly the same amount of iron sheet, 30,800 cm ² , for each container. Do you think they hold the same amount of maize?' Then give 2 minutes of silent writing. Say: 'Write your prediction and your reason. There is no wrong answer today; I want to see your thinking.' Move around and note, without commenting, which learners write 'same material, so same amount'. Use WAIT TIME of 10-15 seconds after each question before taking responses. Cold-call 3 learners with different predictions: 'Wanjiru, what did you predict and what is your reason?' After each answer ask a different learner: 'Who can restate that idea in your own words?' Do not confirm or correct. Say: 'Keep your slip safe. We will come back to it at the end.' Tally the predictions on the board.	Prediction with reasoning and public tallying (eliciting prior ideas). Writing a prediction and a reason makes learners commit to the common intuition that equal material means equal capacity. The class tally makes the disagreement visible, so the puzzle becomes a shared problem and gives a reason to investigate. It also allows the teacher to see how many hold the misconception that surface area and volume are the same thing.	Observe and record: (1) number of learners who predicted 'the same' and gave the reason 'same sheet metal' (indicates the surface area = volume misconception); (2) whether learners distinguish material used (surface) from amount held (inside); (3) whether at least 8 of 10 learners complete both parts (a) and (b) of the prediction slip with a reason. Collect slips from 5 randomly chosen learners to compare with the end-of-lesson revision.
Observe Phase  VIDEO: Surface area using a net: rectangular prism Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Surface Area of	In pairs, learners observe and measure physical prism models and build nets. Task A (rectangular prism): measure a 5 cm x 4 cm x 3 cm cardboard box (or make one), unfold it flat, or draw and cut out its net from cardboard, labelling each face with its dimensions. They identify that there are 6 rectangular faces in 3 equal pairs: 5 x 4 = 20 cm ² (two), 5 x 3 = 15 cm ² (two), 4 x 3 = 12 cm ² (two). They add to get TSA = 2(20 + 15 + 12) = 94	Distribute pre-cut cardboard (reuse old cereal and soap cartons), rulers, pencils and blunt-tip scissors to each pair. Say: 'Wanjiru did not guess. She measured, and she worked with flat pieces of sheet. Today you will do the same: unfold the solid into a flat net and measure every face.' Remind learners of the safety rule: 'Scissors point down and away from you.' Circulate and ask: 'How many faces does your prism have?'	Modelling with physical nets and data tables (Developing and using models). Unfolding a 3D prism into a flat net lets learners see every face at once, so no face is missed or counted twice. The data table organises the areas, and comparing the two prisms helps learners notice the structure: two identical ends plus a set of rectangles that wrap around. This pattern-finding is the basis for the general formula in the Explain	Check each pair's net and table. Indicators: (1) the rectangular prism net has 6 faces labelled with correct areas 20, 20, 15, 15, 12, 12 cm ² ; (2) the triangular prism net shows 2 triangles of 6 cm ² and 3 rectangles of 30, 40, 50 cm ² ; (3) totals of 94 cm ² and 132 cm ² are correct; (4) the learner can say that the rectangles' widths together equal the perimeter of the triangle (3 + 4 + 5 = 12 cm). Mark pairs who forgot the second triangle or

Triangular Prisms Practice Source: CK-12 Search ARES for similar readings	cm2. Task B (triangular prism): make or examine a prism with right-angled triangle ends 3 cm, 4 cm, 5 cm and length 10 cm. They draw its net: 2 triangles (each $\frac{1}{2} \times 3 \times 4 = 6 \text{ cm}^2$) and 3 rectangles ($3 \times 10 = 30$, $4 \times 10 = 40$, $5 \times 10 = 50$, which total 120 cm^2). $\text{TSA} = 12 + 120 = 132 \text{ cm}^2$. They record results in a table with columns: Face, Shape, Dimensions, Area (cm^2). Then they look at both tables and look for a pattern linking the two ends (bases) and the rectangles around them.	Which faces are identical? How do you know?' Use WAIT TIME of 10-15 seconds before accepting answers. Prompt pairs who finish early: 'The two triangles in the net are the bases. What is the total length of the rectangles laid side by side, and how does it compare with the perimeter of the triangle?' Cold-call 2 pairs to report their TSA for each prism: 'Pair at the window, what is your total for the rectangular prism? Pair at the back, what did you get for the triangular prism?' Confirm 94 cm^2 and 132 cm^2 only after the evidence is shared. Draw the table on the board and fill it with learners' values.	phase.	the second set of faces and re-ask the guiding question.
Explain Phase  VIDEO: Surface area of a box (cuboid) Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Surface Area of Rectangular Prisms (Flexbook) Source: CK-12 Search ARES for similar readings	Learners use their tables to generalise. In pairs they answer: What do the two bases contribute? ($2 \times \text{base area}$) What do the rectangles around the base contribute? (together they form one long rectangle of length = perimeter of base and width = length of the prism). They write: $\text{TSA of a prism} = 2 \times \text{base area} + \text{perimeter of base} \times \text{length}$. They check the formula on the $5 \times 4 \times 3$ prism (taking 5×4 as the base: $2 \times 20 + 18 \times 3 = 40 + 54 = 94 \text{ cm}^2$) and on the 3-4-5 prism ($2 \times 6 + 12 \times 10 = 132 \text{ cm}^2$). They then apply it to Mzee Kiplagat's box: square base $50 \text{ cm} \times 50 \text{ cm}$, height 129 cm (the length of this prism). Base area = $2,500 \text{ cm}^2$, so $2 \times \text{base} = 5,000 \text{ cm}^2$. Perimeter = 200 cm, so $200 \times 129 = 25,800 \text{ cm}^2$. $\text{TSA} = 5,000 + 25,800 = 30,800 \text{ cm}^2$, the same $30,800 \text{ cm}^2$ of iron sheet that Wanjiru used. Learners also sketch the net of the box. Finally they write one sentence distinguishing surface area (the	Say: 'Look at your two tables. What stays the same in both? What changes?' Use WAIT TIME of 10-15 seconds, then cold-call 3 learners to describe the pattern in their own words. Build the formula on the board from learner responses, not by telling: 'Two bases, plus the rectangles that wrap around. If I lay the rectangles side by side, how long is that strip?' Write $\text{TSA} = 2 \times \text{base area} + \text{perimeter} \times \text{length}$. Ask: 'Why does perimeter appear in the formula?' Cold-call 2 learners. Then pose the key task: 'Mzee Kiplagat's box is 50 cm by 50 cm by 129 cm. Use the formula to find how much sheet Wanjiru needed.' Let pairs work for 4 minutes, then cold-call 2 pairs to show the steps. Confirm $30,800 \text{ cm}^2$ and say: 'This matches the amount of iron sheet Wanjiru used. So the maths agrees with the real workshop.' Address the misconception: 'We found how much material covers the box. Have we found how much	Pattern generalisation from data followed by application to the anchor (claim-evidence-reasoning). Learners derive the formula from their own measurements, test it on two prisms (evidence), and then use it on the real container (application). Linking the result to Wanjiru's $30,800 \text{ cm}^2$ validates the mathematics against the phenomenon. Explicitly separating 'covering' from 'holding' prepares learners to see why equal surface area does not guarantee equal volume.	Observable indicators: (1) learners write $\text{TSA} = 2 \times \text{base area} + \text{perimeter} \times \text{length}$ correctly; (2) they identify the base of the box as 50×50 and the length as 129 cm; (3) they calculate $5,000 + 25,800 = 30,800 \text{ cm}^2$ with the unit cm^2; (4) they can explain in one sentence why perimeter $\times$ length gives the area of the rectangles around the base; (5) they distinguish surface area from volume in writing. Use a 3-question exit check on mini-whiteboards: (a) TSA of a $6 \text{ cm} \times 5 \text{ cm} \times 2 \text{ cm}$ prism (base 6×5: $2 \times 30 + 22 \times 2 = 104 \text{ cm}^2$); (b) 'Does the box's TSA tell us how much maize it holds? Yes or no, and why?'; (c) 'Which unit do we use for surface area?' Aim for at least 80% correct; revisit the formula with those who did not reach it.

	material covering the outside) from volume (the space inside), and record that the box's volume has not been found yet.	maize it holds? Not yet. Surface area and volume are different quantities.' Do not teach the drum's SA or any volume formula today.		
Driving Question Board (DQB) Creation  VIDEO: Surface area using a net: triangular prism Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Surface Area and Volume of Prisms Practice Source: CK-12  Search ARES for similar readings	The teacher displays the driving question on a large chart: 'Why does the same amount of metal sheet make containers that hold very different amounts of grain, and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction?' Learners individually write at least one question on sticky notes or paper strips about the phenomenon (for example: How much does the drum hold? What is the drum's surface area? How do we find the volume of a cylinder? Is there a shape that holds the most? Why would a round container be better?). In groups of four they sort their questions into categories such as Surface area questions, Volume questions, and Shape and design questions, and post them on the DQB. Beside the board they add a small 'What we know so far' panel: the box's TSA is 30,800 cm²; TSA of a prism = 2 x base area + perimeter x length; surface area is not the same as volume.	Say: 'Scientists and engineers begin with questions. What are you still wondering about Mzee Kiplagat's containers?' Give 3 minutes of silent writing and then use WAIT TIME of 10-15 seconds before groups sort. Circulate and push for deeper questions: 'Can you turn that into a question we can investigate with measurements?' Cold-call 4 learners to read one question each and place it on the board with a reason for its category. Highlight questions that match upcoming lessons, for example: 'Wekesa asks how to find the drum's surface area. We will investigate curved surfaces later in this unit.' Do not answer volume questions now; say: 'Good question. Let us keep it on the board and come back with evidence.' Point to the KICD key inquiry questions and link them to learners' questions. Keep the DQB visible for all remaining lessons.	Driving Question Board (student-generated questions). The DQB gives learners ownership of the unit's direction and keeps the phenomenon at the centre. Sorting questions into categories shows how the lessons connect, and the 'What we know so far' panel records the first piece of evidence (box TSA = 30,800 cm²) so progress is visible across lessons.	Indicators: (1) every learner contributes at least one question; (2) at least 3 groups include a question about volume or capacity, which shows they recognise that volume is a separate idea; (3) at least 2 questions ask about the drum or about shape (for example, why round?), showing connection to the phenomenon; (4) learners place questions in sensible categories and justify their choice. Photograph or copy the DQB so questions can be compared at the end of the unit.
Model Building Phase  VIDEO: Surface area using a net: rectangular prism Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners start the class's first version of the explanatory model on a chart titled 'Why does the drum hold more maize?'. Individually they sketch the two containers (drum and box) with labels: both closed, both 30,800 cm² of sheet. Under the box they write the dimensions 50 cm x 50 cm x 129 cm and 'TSA = 30,800 cm²'. Under the drum they write the dimensions $r = 35$ cm and $h =$	Say: 'Today your model is a first draft. We will keep adding to it in every lesson until we can explain why the drum holds more.' Give learners 5 minutes to draw and write, then use WAIT TIME of 10-15 seconds before sharing. Cold-call 3 learners to read their one-sentence explanation. Ask: 'What do we know for certain? What do we still need to find?' (We know the box's TSA is 30,800 cm²; we	Cumulative model building (initial model with open questions). Drawing the two containers with known and unknown values shows exactly what has been established and what remains. Writing the distinction between surface area and volume in the model makes the central idea explicit, and revising the earlier prediction shows learners how their thinking has grown. The model will be	Check each model for: (1) both containers drawn with correct dimensions (drum $r = 35$ cm, $h = 105$ cm; box 50 cm x 50 cm x 129 cm); (2) box TSA written as 30,800 cm² with the correct unit; (3) surface area and volume labelled as different quantities; (4) at least one open question about the drum or volume; (5) a revised prediction sentence. Learners who still say 'same sheet, so same maize'

 HTML: Surface Area of Prisms (Flexbook) Source: CK-12 Search ARES for similar readings	105 cm and a question mark for TSA and volume, because these are not yet calculated. They add an arrow labelled 'material that covers = surface area' and another labelled 'space inside = volume' and write a first explanation in their own words, for example: 'Surface area tells how much sheet is used. Volume tells how much maize it holds. We need to compare them.' Learners revisit their prediction slips, put a tick or a cross, and write one sentence about whether their thinking has changed. They complete the summary table (see below) in their books.	need the drum's SA, and both volumes.) Ask learners to hand in or keep their models in their folders. Give the closing prompt: 'Next lesson we will look at roofs on a school hall in Kajiado and learn how to find surface area when the faces are triangles that slope. As you walk home, look at a water tank or a sufuria and ask yourself: is it the covering or the contents that matters?' Remind them to return all scissors and cardboard offcuts to the collection box.	revised after each lesson, ending in Lesson 8.	should be seen in the next lesson's starter. Record the number of learners meeting 4 or more of the 5 indicators.
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D. TEACHER REFLECTION

After the lesson ask yourself: (1) How many learners still believed at the end that equal surface area means equal capacity, and what evidence from the prediction slips and models shows this? (2) Did all pairs build both nets correctly, and which faces were most often missed (usually the second base or the second set of rectangles)? (3) Could learners explain in their own words why perimeter x length gives the area of the rectangles around the base? (4) Did the DQB produce questions about volume and about shape, and did I protect those questions from being answered too early? (5) Which learners need extra support with area of rectangles and triangles from Grade 9 before Lesson 2? (6) Were materials, such as cardboard and scissors, sufficient and safely used? Plan adjustments for the next lesson, for example a quick starter on Pythagoras and slant height.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed two closed containers at Mzee Kiplagat's posho mill in Eldoret, a cylindrical drum and a square-based box, both made from 30,800 cm ² of iron sheet and both about 1 m tall, yet the drum holds more maize. We also measured the faces of a 5 cm x 4 cm x 3 cm rectangular prism (TSA 94 cm ²) and a 3-4-5 cm triangular prism of length 10 cm (TSA 132 cm ²).
What did I learn?	The total surface area of a prism is the sum of the areas of all the faces in its net: $TSA = 2 \times \text{base area} + \text{perimeter of base} \times \text{length}$. The box (50 cm x 50 cm x 129 cm) has $TSA = 5,000 + 25,800 = 30,800 \text{ cm}^2$. Surface area measures the material covering a solid, and volume measures the space inside, so they are different quantities.
How does this explain the phenomenon?	Wanjiru used 30,800 cm ² of sheet for the box, which agrees with our calculation, so the same amount of material is used for both containers. But knowing the surface area does not tell us how much maize a container holds, because capacity depends on volume. We still need to find the drum's surface area and both volumes to explain why the drum holds more; this is now on our DQB.

LESSON 2 (80 minutes (2 x 40-minute periods)): Surface Area of Pyramids: Slant Height, Vertical Height and Roof Cover

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners build on Lesson 1 (nets and the sum-of-faces method for prisms) to find the surface area of pyramids, and they distinguish slant height (l) from vertical height (h). This advances the driving question by showing that Wanjiru, the jua kali metalsmith, must use the correct measurement (slant height, not vertical height) to calculate the sheet or roofing material a solid needs. It also shows that surface area measures covering material only.
Knowledge	By the end of the lesson the learner should be able to: (a) distinguish the slant height (l) of a pyramid from its vertical height (h); (b) use Pythagoras' theorem to find l from h and half the base side ($3^2 + 4^2 = 5^2$); (c) state and apply pyramid TSA = base area + sum of the areas of the triangular faces; (d) calculate the roof area of a square-based pyramid roof (6 m base, h = 4 m, l = 5 m, roof area = $4 \times \frac{1}{2} \times 6 \times 5 = 60 \text{ m}^2$) and the surface area of a regular triangular pyramid (tetrahedron, all edges 6 cm, each face $9\sqrt{3} \approx 15.59 \text{ cm}^2$, TSA $\approx 62.35 \text{ cm}^2$).
Skills	Constructing and folding nets (Developing and Using Models); measuring and labelling; applying Pythagoras' theorem; analysing and interpreting data (comparing measurements of h and l); using mathematics and computational thinking; constructing explanations; communicating and collaborating in groups; critical thinking and problem solving.
Attitudes	Responsibility: learners take care of the materials and use cut-outs from the immediate environment carefully. Integrity: learners measure and record accurately and report their true measurements. Unity: learners work together to build the nets and share their ideas. Appreciation of how mathematics is used in construction and roofing in Kenya.
Key Inquiry Question	How do we determine the surface area and volume of solids? How are surface area and volume of solids applied in real-life situations?
Purpose in Storyline	Lesson 1 launched the Eldoret phenomenon: the drum and the box both use 30,800 cm ² of iron sheet, and the box's SA was found with nets and the sum-of-faces method. Lesson 2 extends the same net idea to pyramids. It introduces slant height, which Lesson 3 will need for cones and frustums, and it reinforces that surface area is only the covering material, so it cannot tell us how much a solid holds. That prepares learners for the volume lessons (L5 and L6) that resolve why the drum holds more maize than the box.
Safety Notes	Scissors and craft knives: cut away from the body and keep fingers clear of the blade. Do not climb onto any real roof or structure; take all measurements from models, photographs and the diagram. Keep the floor clear of paper offcuts to prevent slips. Do not leave scissors on the edges of desks. Use blunt-tipped scissors where possible.

B. LESSON OVERVIEW

Mzee Kiplagat has since asked Wanjiru to roof a small store at his posho mill with a pyramid-shaped iron-sheet roof, just like the school hall roofs and Maasai manyatta-style roofs seen in Kajiado and Narok. Learners first predict how much roofing sheet a 6 m x 6 m pyramid roof with a vertical height of 4 m will need. They then construct nets of a square-based pyramid and a regular triangular pyramid (tetrahedron), measure and compare vertical height with slant height, and use Pythagoras to find l = 5 m. They derive that pyramid TSA = base area + areas of the triangular faces, and calculate that the roof area is $4 \times \frac{1}{2} \times 6 \times 5 = 60 \text{ m}^2$ (the triangular faces only, since a roof has no floor). A tetrahedron with edges of 6 cm gives about 62.35 cm². Learners revisit the Driving Question Board and add pyramid evidence to the class model: surface area is the covering material needed, found by adding the face areas, and it is a different quantity from volume.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Surface area word problem example Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Surface Area and Volume of Pyramids Practice Source: CK-12  Search ARES for similar readings	Learners look at photographs of a Maasai manyatta roof from Narok and a school hall roof in Kajiado. Mzee Kiplagat wants Wanjiru to roof his store with a square-based pyramid roof with a 6 m x 6 m base and a vertical height of 4 m. In pairs, learners individually write then share a prediction: How many m² of iron sheet will cover the four sloping faces? They also say which length of the roof they would use for the height of one triangular face, and why. Some will predict $4 \times \frac{1}{2} \times 6 \times 4 = 48$ m² using the vertical height; others will guess something else. Learners record their prediction in their books, with a reason, and draw a quick sketch of the roof showing the length they think they need. They link this back to Lesson 1: 'we found the SA of the box by adding the faces, so can we do the same here?'	Display the two photographs and say: 'Wanjiru has to buy iron sheet for this roof. Before we learn anything new, what do you think she needs to know?' Give individual silent think time and WAIT TIME of 10-15 seconds before anyone speaks. Then ask learners to turn to a partner for 2 minutes. Cold-call 4 pairs to share their predictions, and write each one on the board without judging it. Ask: 'Which length did you use for the height of a triangular face, and why?' Ask: 'What did we do in Lesson 1 to find the surface area of the box, and could that help here?' Do not say whether any prediction is correct. Keep the predictions displayed for later comparison. Then ask the whole class: 'Do we expect the roof to need as much sheet as the drum or box used?' to link it back to 30,800 cm².	Predict-Observe-Explain (Prediction) with Think-Pair-Share: Recording a prediction and a reason before instruction exposes learners' ideas (for instance, that vertical height can be used for the triangular face). When learners later compare their prediction with evidence, they experience a conflict that motivates the idea of slant height, and they own the correction.	Look for: (1) how many learners use the vertical height of 4 m in their calculation, a sign of the slant-height misconception, and tally them on the board; (2) whether learners suggest adding the areas of the faces, which shows transfer from Lesson 1; (3) whether any learners confuse 'area of the roof' with 'space inside the roof' (volume). Record the numbers of each type of response. Use a thumbs-up/down check for 'Is it possible to know the roof area without measuring a sloping edge?'
Observe Phase  VIDEO: Surface area word problem example Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Surface Area and Volume of Pyramids (PLIX) Source: CK-12  Search ARES for similar readings	In groups of 4, learners use a pre-drawn net (or draw their own on manila paper or old cartons) to build two models: (A) a square-based pyramid with a 6 cm square base and four triangular faces (scale model of the 6 m roof, where 1 cm = 1 m, vertical height 4 cm), and (B) a regular triangular pyramid (tetrahedron) with all edges 6 cm. Learners cut, fold and tape the nets. For model A, they first draw the net, measure a triangular face, and then measure the vertical height (from the apex straight down to the centre of the base) with a ruler and a straight stick, and the slant height (from the apex down the middle of a triangular face to the midpoint of a	Distribute nets, rulers, scissors, tape and sticks. Say: 'Your job is to find the length that Wanjiru must use for the height of each triangular face. Measure, do not guess.' Circulate and ask each group: 'Which line is the vertical height and which is the slant height? Show me with your finger on the model.' Hold up a model and say: 'Where does the slant height touch the base?' Pose the challenge: 'Why is the slant height longer than the vertical height?' WAIT TIME of 10-15 seconds, then cold-call 3 groups. Prompt with: 'Which right-angled triangle can you see inside the pyramid? What are its three sides?' Remind learners about safety with	Model-based investigation with data table (Developing and Using Models; Analysing Data): Building the physical model and measuring both h and l from the same solid makes the difference visible and concrete. Organising results in a table lets learners see patterns (area of one face x number of faces) before they are given a formula, building understanding from evidence rather than from rote.	Observable indicators: (1) the learner can point to the vertical height inside the pyramid and to the slant height along a face and say why they differ; (2) the learner measures l along the face, not along the edge of the pyramid; (3) the learner records $l = 5$ for model A and checks it with $3^2 + 4^2 = 5^2$; (4) the learner calculates a single triangular face area using l as the height of the triangle. Use a checklist of these four items for each group. Intervene if a group uses the edge length instead of slant height, and ask them to compare with a right-angled triangle drawn on the net.

	base edge). They record $h = 4$ and $l = 5$ (scale units) and check that $3^2 + 4^2 = 5^2$ (the right-angled triangle formed by the vertical height, half the base side, and the slant height). They then calculate the area of one triangular face ($1/2 \times 6 \times 5 = 15$), then four faces (60). For model B, they measure the base side (6 cm) and the height of one face (about 5.2 cm), and compute each face as $9\sqrt{3} \approx 15.59 \text{ cm}^2$. Groups record their measurements in a table with columns: solid, base, h, l, area of one face, number of faces, total.	scissors. Remind learners that, as in Lesson 1, the net shows every face laid flat. Do not give the formula yet. Ask groups to put away their tapes only after they have recorded the data. Then ask: 'Is the base counted if this is a roof?' and 'Is it counted if we were making a closed pyramid-shaped container?'		
Explain Phase  VIDEO: Surface area using a net: triangular prism <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Surface area word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Using their measurements, learners explain to their groups and then to the class: (1) why the height of each triangular face is the slant height l and not the vertical height h; (2) how to find l when only h and the base side are known, using Pythagoras: $l^2 = h^2 + (\text{half the base})^2 = 4^2 + 3^2 = 25$, so $l = 5 \text{ m}$; (3) how to derive the formula. They fill in the sentence frame: 'Pyramid TSA = base area + the sum of the areas of the triangular faces.' For the roof, they decide that the base (the floor of the store) is not covered with iron sheet, so the roof area = $4 \times 1/2 \times 6 \times 5 = 60 \text{ m}^2$, while the closed-pyramid TSA would be $36 + 60 = 96 \text{ m}^2$. For the tetrahedron, they calculate each face as $9\sqrt{3} \approx 15.59 \text{ cm}^2$ and the TSA as $4 \times 15.59 \approx 62.35 \text{ cm}^2$. Learners then solve a new problem: a pyramid-shaped roof for a school hall in Kajiado with a $8 \text{ m} \times 8 \text{ m}$ base and a vertical height of 3 m; find $l = 5 \text{ m}$, roof area = $4 \times 1/2 \times 8 \times 5 = 80 \text{ m}^2$. Learners write a short statement correcting their earlier prediction and say what the	Ask: 'Using what you measured, why can't we use 4 m as the height of the triangle?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners, and push for evidence: 'What did you measure that shows this?' Draw the right-angled triangle ($h = 4$, half-base = 3, $l = 5$) on the board and have learners come up and label it. Say: 'Slant height is never called height. We write l for slant and h for vertical.' Introduce the formula only after learners propose it in their own words. Work the 60 m^2 example on the board with learners calling out each step, then have them solve the $8 \text{ m} \times 8 \text{ m}$, $h = 3 \text{ m}$ school hall roof in pairs. Cold-call 4 pairs for l, a face area, and the total. Return to the Predict phase: 'Whose prediction was closest? What was the error in the 48 m^2 prediction?' Ask: 'Wanjiru's two containers had equal surface area. If this roof and a box also had equal surface area, would they hold the same amount?' to prepare for later volume lessons, without teaching volume now. Note the unit conventions: area in m^2,	Claim-Evidence-Reasoning with worked-example fading: Learners make a claim (the height of a triangular face is l), support it with their measurements, and reason using Pythagoras. A fully worked example (60 m^2) is followed by a partly guided problem (80 m^2), which gradually shifts responsibility to the learner and builds procedural fluency along with conceptual understanding.	Collect an exit-style written response to: 'Find the roof area of a square pyramid with base 6 m, $h = 4 \text{ m}$. Show l.' Success criteria: finds $l = 5 \text{ m}$ by Pythagoras (or states it), uses $1/2 \times 6 \times 5 = 15 \text{ m}^2$ per face, multiplies by 4 to give 60 m^2, writes the unit as m^2, and states that the base is excluded for a roof. Check the Kajiado hall answer (80 m^2). Tally how many learners still use h instead of l. Use the common errors to plan a re-teach at the start of Lesson 3, where slant height is needed again.

	answer of 60 m ² tells us about the material needed, and what it does not tell us (how much space is inside the roof).	never m or m ³ .		
Driving Question Board (DQB) Creation  VIDEO: Surface area of a box (cuboid) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Surface Area and Volume of Pyramids Practice Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board started in Lesson 1. Each group writes on sticky notes (a) one new thing they learned (for example, 'a pyramid's triangular faces use the slant height, not the vertical height'), and (b) one new question that arose (for example, 'Does the 60 m ² roof also tell us how much space is inside?', 'If the box were a pyramid, would it hold more than the drum?', 'What will we need for a cone, which has a curved surface?'). They place the notes under the headings: 'What we have found', 'What we still need to find out'. They link at least one note to the phenomenon: surface area is only the covering material, so it cannot explain why the drum holds more than the box with the same 30,800 cm ² .	Say: 'Take two minutes to write what you now know and what you still wonder.' Give WAIT TIME of 10-15 seconds before groups begin. Ask each group to read out one note, and cold-call 5 groups. Group the notes with the class: 'Which of these are about surface area, and which are about volume or space inside?' Point to the Eldoret photos: 'Does the pyramid roof help us explain why the drum holds more maize than the box?' Elicit that it shows how to measure covering material for flat faces, but not capacity. Highlight questions to be answered in Lessons 3 and 4 (curved surfaces) and Lessons 5 and 6 (volume). Do not answer the volume questions now; mark them as 'coming soon'.	Driving Question Board with question sorting: Learners' own questions are sorted by type (surface area vs volume, flat vs curved). Seeing how their questions relate to the anchoring phenomenon maintains a purpose for learning and shows progress, as questions about pyramids are answered while new ones point to cones, spheres and volume.	Check the sticky notes for: (1) correct use of the terms slant height and vertical height; (2) whether learners distinguish surface area (covering) from volume (space inside); (3) whether at least one question is linked to the drum and box. Count how many groups produce a volume question, which shows readiness for Lessons 5 and 6. Note any notes that call slant height 'height' and address them in the next lesson.
Model Building Phase  VIDEO: Surface area using a net: rectangular prism <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Surface Area and Volume of Pyramids (PLIX)	Learners revise the class model (a poster or diagram) that is being built from Lesson 1 to answer the driving question. They add a 'Pyramids' section containing: a labelled sketch of a square pyramid showing h, l, half-base and the right-angled triangle; the rule 'Pyramid TSA = base area + sum of triangular faces'; the worked figures (roof 60 m ² ; tetrahedron about 62.35 cm ²); and a note: 'Surface area tells us how much material covers a solid, not how much it holds.' In their notebooks they complete a three-column summary (What we observed, What we learned, How it helps explain the drum and the	Display last lesson's model and say: 'What does our model say so far about the drum and the box?' (the box's SA is 30,800 cm ² , found by adding the faces of its net). Then ask: 'What is new today that we should add?' Give WAIT TIME of 10-15 seconds and cold-call 3 learners. As groups draw, circulate and check that l and h are labelled correctly and units are m ² or cm ² . Ask: 'Where on our model do we show that the same sum-of-faces method works for prisms and pyramids?' Ask a volunteer to read the last line aloud and ask: 'Does our model explain yet why the drum holds more maize?' Guide learners to say that it does not yet,	Cumulative model revision with gallery review: Learners add to a single evolving model across lessons, so each new idea (slant height, sum-of-faces for pyramids) is connected to earlier ones (nets, prisms) and to the phenomenon. Writing sentence frames in their own words consolidates the explanation, and noting what is still missing (volume) creates a need for the next lessons.	Review the model posters and notebook summaries for: (1) correct labelling of h, l and the half-base in the right-angled triangle; (2) the correct statement of the rule and figures (60 m ² ; about 62.35 cm ²); (3) a sentence showing that surface area differs from volume; (4) a connection to the drum and the box. Use a 3-level check (secure, nearly, not yet) and note learners who need extra support with Pythagoras or with identifying l. Plan a short starter in Lesson 3 to revisit slant height with those learners.

Source: CK-12  Search ARES for similar readings	box) and write two sentences that begin: 'Wanjiru would use the slant height because...' and 'This helps explain the drum and the box because...'. They also add a 'careers' note: quantity surveyors and roofers in Kenya use these calculations to order iron sheets.	and that we need volume. Collect the notebooks' summary tables for review. Close by saying: 'Next lesson, we meet curved surfaces, and slant height will return.'		
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D. TEACHER REFLECTION

After the lesson, consider: (1) How many learners used the vertical height (4 m) in their first prediction, and how many moved to the slant height (5 m) after measuring? Did the physical model make the difference clear? (2) Could learners find l by Pythagoras on their own, or did they need the right-angled triangle drawn for them? (3) Did any learners confuse roof area (4 faces, 60 m²) with the total surface area of a closed pyramid (96 m²)? How could l make the 'is there a floor?' question clearer? (4) Did learners clearly keep surface area (material) and volume (space inside) separate in their discussion? (5) Which pairs and groups participated least, and how can l involve them in Lesson 3? (6) Was the 80-minute time enough for building the nets and the practice problem? What should be adjusted next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We built a square-based pyramid (base 6 m, vertical height 4 m, scale model) and a regular triangular pyramid (tetrahedron, edges 6 cm). We measured the vertical height and the slant height and saw that the slant height (5 m) is longer than the vertical height (4 m). We saw that the triangular faces need the slant height.
What did I learn?	Slant height (l) is measured along a triangular face; vertical height (h) goes straight down from the apex to the base. We can find l using Pythagoras ($3^2 + 4^2 = 5^2$). Pyramid TSA = base area + sum of the areas of the triangular faces. The roof area = $4 \times \frac{1}{2} \times 6 \times 5 = 60$ m ² , and the tetrahedron has TSA of about 62.35 cm ² .
How does this explain the phenomenon?	Wanjiru can use the net and sum-of-faces method to find how much iron sheet any flat-faced solid needs, and she must use the slant height for the triangular faces. Surface area only measures the covering material, not the space inside, so it cannot yet explain why the drum and box with the same 30,800 cm ² hold different amounts of maize. We need to study volume to answer that.

LESSON 3 (40 minutes (single lesson); extend to 80 minutes if double period): Surface Area of Cones and Frustums: From Sector Nets to Turkana Water Tanks

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners derive the curved surface area of a cone (πrl) from its sector net and extend it to the frustum (cone minus removed cone, $\pi(R+r)l$). They use these to calculate the sheet metal needed for a conical tank top and the Turkana frustum tank, adding curved-surface tools to the evidence for why shape and surface area are different from capacity in the drum-and-box puzzle.
Knowledge	By the end of the lesson the learner should be able to: (a) identify the slant height l and vertical height h of a cone and relate them using $l^2 = r^2 + h^2$ (building on Lesson 2: slant height vs vertical height and Pythagoras for l); (b) explain that the net of a cone's curved surface is a sector of a circle whose arc equals the circumference of the base, and derive curved SA = πrl ; (c) state that total SA of a cone = $\pi r(r + l)$, always saying whether the base is included; (d) derive the frustum's curved SA as big cone minus removed small cone = $\pi(R + r)l$; (e) calculate the ice-cream cone ($r = 3.5$ cm, $l = 12$ cm, curved SA = 132 cm ²), the conical tank top ($r = 1.4$ m, $l = 2$ m, 8.8 m ²) and the Turkana frustum (17.325 m ²). Use $\pi = 22/7$.
Skills	Using NGSS-aligned practices: Developing and Using Models (building and unrolling a cone net, representing a frustum as a big cone minus a small cone); Analyzing and Interpreting Data (comparing measured and calculated areas); Using Mathematics and Computational Thinking (deriving and applying formulae, Pythagoras); Constructing Explanations (justifying why the sector's radius is l and not r); Engaging in Argument from Evidence (checking the frustum result two ways). Core competencies: Critical Thinking and Problem Solving; Communication and Collaboration; Creativity and Imagination.
Attitudes	Values: Responsibility (taking care of the materials and measuring carefully); Integrity (reporting measured values honestly, even when they do not match perfectly); Unity (working together in groups). PCI: Education for Sustainable Development (reusing scrap card and paper, and minimising wasted sheet metal, as a metal fabricator would). Appreciate the use of surface area calculations by artisans in real-life situations.
Key Inquiry Question	1. How do we determine the surface area and volume of solids? 2. How are surface area and volume of solids applied in real-life situations?
Purpose in Storyline	Lessons 1 and 2 gave learners the sum-of-faces method for flat-faced solids (prisms and pyramids) and the slant height idea; the box's SA of 30,800 cm ² is already established. Lesson 3 adds the first curved-surface formulae, which Lesson 4 will use for the sphere and the drum's total SA check. The lesson also introduces the Turkana frustum tank, which Lesson 6 will use for volume. For the driving question, learners see how an artisan like Wanjiru works out how much sheet is needed for round, tapering shapes, preparing them to understand why the drum's circular cross-section matters. Lesson 3 does NOT yet compute volumes or resolve the drum vs box puzzle; those come in Lessons 5 and 6.
Safety Notes	Scissors and cutting card: cut away from the body and keep fingers clear. Use compasses and pencil points carefully and keep them capped or stored when not in use. Do not use real sharp metal sheet in the classroom; use card or paper only. Keep walkways clear of paper off-cuts to prevent slips. Real ice-cream cones should only be handled if learners have clean hands and no allergy to the contents; a cardboard or paper cone is a safe alternative.

B. LESSON OVERVIEW

Mzee Kiplagat's two containers each used 30,800 cm² of Wanjiru's iron sheet, and Lesson 1 showed that the box's faces really do add up to that figure. But the drum is

round, and its curved wall is not made of flat rectangles you can count. In this lesson learners study another round shape, the cone, using an ice-cream cone, a conical heap of maize or ugali flour, and the conical top of a water tank. They predict how much paper is needed to cover a cone, then cut open a paper cone to see its net: a sector of a circle (extending the net idea from Lesson 1). Measuring the sector shows that its radius is the slant height l and its arc is the base circumference, giving curved $SA = \pi rl$. Learners then use the conical tank-top data ($r = 1.4$ m, $l = 2$ m, 8.8 m²) and extend to the Turkana frustum: a big cone ($R = 2.1$ m, slant 3.5 m) with a small cone ($r = 1.05$ m, slant 1.75 m) removed, giving $\pi(R + r)l = 17.325$ m², checked by $23.1 - 5.775$. The DQB and model-building phases connect the new curved-area tools to the puzzle: SA tells us how much metal we need, and we still need volume to know the capacity.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Volumes of cones intuition <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Surface Area and Volume of Cones Practice Source: CK-12  Search ARES for similar readings	(About 6 minutes.) Learners sit in groups of 4. Each group receives a cardboard or paper cone, a sheet of plain paper, and a ruler. The teacher shows an ice-cream cone ($r = 3.5$ cm, slant $l = 12$ cm) alongside a photo of the conical roof of a Maasai manyatta in Narok and the conical top of a water tank. The teacher recalls Wanjiru's problem: she must know how much sheet to cut. Individually, each learner writes in their notebook: (1) A prediction: 'If I cut the cone open along its slant edge and flatten it, what shape will the paper make?' with a sketch. (2) A guess for the sheet needed for the ice-cream cone's curved part, in cm ² , with a reason. Then groups compare predictions and agree one group prediction on a mini-whiteboard or page. Typical learner predictions: a triangle, a rectangle, a circle, a half-circle, or a 'slice of pizza'. Learners are also asked to say whether they think the base should count in the 'sheet needed' and why.	1) Hold up the ice-cream cone and say: 'Wanjiru made a drum and a box from the same sheet. Today we ask: how would she cover a cone? Before we do anything, I want your honest prediction.' 2) Give 2 minutes of silent individual writing, then say: 'Write your sketch first; do not look at your neighbour.' 3) Ask: 'Which part of this cone is flat and which part is curved?' and use WAIT TIME of 10-15 seconds before taking answers; cold-call 3 learners. 4) Ask: 'In Lesson 2 we met slant height. Where is the slant height on this cone, and is it the same as the vertical height?' Wait 10 seconds, cold-call 2 learners to point it out with a ruler or finger. 5) Circulate and note, without correcting, the predicted net shapes: record tallies for 'triangle', 'rectangle', 'circle/part of circle' on the board. 6) Say: 'There is no wrong prediction today. We will keep these and test them with evidence.' Do not reveal the sector yet.	Predict-Observe-Explain (POE) with individual commitment before group talk: writing a prediction and a sketch gives each learner a concrete idea to test, and a tally of competing predictions creates a shared reason to investigate. Linking to the flat nets from Lesson 1 (rectangles and triangles) lets learners notice that a curved surface does not obviously unroll into a familiar shape, creating productive curiosity.	Observable indicators: (a) Learner sketches show a net shape and not just a solid; (b) learner can point to slant height versus vertical height on the cone (listen for 'l is along the sloping side'); (c) predictions include a stated reason, not only a shape. Teacher tally of predicted net shapes on the board (count how many groups say triangle, rectangle, circle part). If more than half of learners point to the vertical height when asked for slant height, plan a quick Pythagoras reminder ($l^2 = r^2 + h^2$) from Lesson 2 before the next phase. Collect 3 sample predictions to compare later.
Observe Phase  VIDEO: Volumes of cones intuition	(About 12 minutes.) Station-style hands-on investigation in the same groups. Task A (cone net): Each group cuts a paper cone open along its slant edge from the base	1) Distribute scissors, string, rulers, and paper cones; remind: 'Cut away from your body, and cut only along the slant edge.' 2) Say: 'Before you cut, hold your	Modelling with a physical net and a geometric argument: unrolling the cone converts an unfamiliar curved surface into a familiar flat sector, then the fraction-of-a-circle	Observable indicators: (a) learner labels the sector's straight edge as l , not r or h ; (b) learner states that the arc of the sector equals the circumference of the base ($2\pi r$),

<i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Surface Area and Volume of Cones Practice Source: CK-12 Search ARES for similar readings	circle to the apex and flattens it. They see a sector of a circle. They measure (i) the straight edge of the sector, which is the slant height l, and (ii) the arc length using string laid along the arc, then compare it to the string length around the base circle of the cone, which is the circumference $2\pi r$. They record in a table: r, l, arc length, base circumference. For the ice-cream cone model they use $r = 3.5$ cm and $l = 12$ cm. Task B (derivation): Using the whole circle of radius l (circumference $2\pi l$) and the sector's arc $2\pi r$, the group reasons that the sector is the fraction $(2\pi r)/(2\pi l) = r/l$ of the whole circle. Area of whole circle = πl^2, so sector area = $(r/l) \times \pi l^2 = \pi r l$. Groups then compute the ice-cream cone: curved SA = $22/7 \times 3.5 \times 12 = 132$ cm². Task C (frustum): The teacher shows a card cone with the top sliced off parallel to the base. Groups lay the pieces together to see that the frustum equals the big cone minus the small cone. Learners record the Turkana tank data: big cone $r = 2.1$ m, slant 3.5 m; small cone $r = 1.05$ m, slant 1.75 m. They calculate: big cone curved SA = $22/7 \times 2.1 \times 3.5 = 23.1$ m²; small cone = $22/7 \times 1.05 \times 1.75 = 5.775$ m²; frustum curved SA = $23.1 - 5.775 = 17.325$ m².	prediction. After you flatten it, compare with the tally on the board.' 3) Circulate and ask each group: 'Which edge of the sector is the slant height? How do you know?' WAIT TIME 10 seconds; if stuck, ask: 'Where was that edge on the cone before cutting?' 4) Ask: 'How does the arc length of your sector compare with the string around the base of the cone?' Wait 10-15 seconds; cold-call 3 groups to report their two numbers. Expect close agreement with measurement error of a few mm; praise honest reporting (Integrity). 5) Draw one large sector on the board and ask: 'The sector is a fraction of a full circle of radius l. What fraction?' Wait 15 seconds, take 2 cold-calls, then write sector fraction = arc $\div$ whole circumference = $2\pi r \div 2\pi l = r/l$. 6) Ask: 'So what is the sector area?' Let groups compute $(r/l) \times \pi l^2$ before revealing $\pi r l$. 7) Introduce the frustum with a card cone: 'Wanjiru cut the top off a Turkana tank cone. How could we find the sheet for what is left?' Wait 10 seconds, cold-call 2 learners to propose 'big minus small'. 8) Check the slant height of the small cone by asking: 'If h and r of the small cone are exactly half of the big one, what is its slant height?' (1.75 m since all lengths halve). Do not re-teach Pythagoras; just ask learners to confirm with $l^2 = r^2 + h^2$ if they want: big cone $h = 2.8$, $r = 2.1$, $l = 3.5$.	argument (r/l of πl^2) builds the formula $\pi r l$ from meaning rather than memory. The frustum is treated by 'whole minus part' reasoning, a decomposition strategy that learners already practised with nets in Lesson 1. Two-way checking (direct formula and subtraction) builds confidence in the result.	with measurements within about 0.5 cm; (c) learner writes the fraction r/l or the equivalent and reaches $\pi r l$; (d) learner computes 132 cm² for the ice-cream cone and 17.325 m² for the frustum without mixing units (cm² vs m²); (e) learner says 'curved' and does not add the base unless told 'total'. Use a 3-finger check: ask learners to show 1, 2, or 3 fingers for their confidence in the derivation; follow up with those showing 1 finger. Note groups that use r in place of l in the formula for the sector's radius.
Explain Phase  VIDEO: Volumes of cones intuition <i>Related topic: not an exact match</i>	(About 10 minutes.) Learners apply the new formulae to a design task. Task 1 (conical tank top): Wanjiru is asked to make a conical top for a tank at a school in Narok with base radius $r = 1.4$ m and	1) Pose Task 1: 'Wanjiru has a roll of sheet. How many square metres does she cut for this conical top?' Give 2 minutes of silent work, then pairs compare. 2) Wait 10-15 seconds after each question; cold-	Claim-Evidence-Reasoning (CER) together with worked contrast: placing 'curved' and 'total' answers side by side (132 vs 170.5 cm²), and the frustum computed two ways (subtraction and formula),	Observable indicators: (a) learner gets 8.8 m² with correct unit and uses $l = 2$ m; (b) learner distinguishes curved (132 cm²) from total (170.5 cm²) and says which one a question needs; (c)

for this lesson. Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Apply volume of solids <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	slant $l = 2$ m. Learners calculate the sheet needed: curved SA = $22/7 \times 1.4 \times 2 = 8.8$ m². They explain in writing why the answer is a curved SA (no base is needed since the cone sits on the tank), and why l, not the vertical height, appears in the formula. Task 2 (total SA contrast): For the ice-cream cone ($r = 3.5$ cm, $l = 12$ cm), a group computes the total SA = $\pi r(r + l) = 22/7 \times 3.5 \times (3.5 + 12) = 11 \times 15.5 = 170.5$ cm², which is $132 + 38.5$ (base circle: $22/7 \times 3.5^2 = 38.5$). Learners state clearly 'curved' versus 'total'. Task 3 (frustum formula): Learners derive $\pi(R + r)l$ from the subtraction. For the Turkana frustum ($R = 2.1$, $r = 1.05$, $l = 1.75$), they compute $22/7 \times 3.15 \times 1.75 = 17.325$ m², matching the $23.1 - 5.775$ result. They explain why the slant height used is 1.75 m, the length along the frustum's slanted side (not the big cone's 3.5 m and not the vertical height 1.4 m). Learners then write a 3-sentence claim-evidence-reasoning response to: 'How does the artisan use mathematics to find the sheet needed for round, tapering shapes?'	call 4 learners across the room for answers and reasoning, not just numbers. Ask: 'Which measurement did you use, and why not the height?' 3) Spot-check for the common error of using $h =$ vertical height; say: 'In the formula, l is the length down the sloping side, measured on the surface.' Use Lesson 2's slant vs vertical idea only as a pointer, not a re-teach. 4) For Task 2, say: 'Say whether your answer is curved or total, out loud, before you write the unit.' Cold-call 2 learners to distinguish 132 and 170.5 cm². 5) For Task 3, draw the frustum cross-section and mark $R = 2.1$, $r = 1.05$, $l = 1.75$ on the board; ask: 'Where is the frustum's slant height on this picture?' Wait 10 seconds; cold-call 3. Then ask learners to expand $23.1 - 5.775$ as $\pi \times 2.1 \times 3.5 - \pi \times 1.05 \times 1.75$ and show $3.5 = 2 \times 1.75$ and $2.1 = 2 \times 1.05$, so the expression becomes $\pi(2 \times 1.05 \times 2 \times 1.75 - 1.05 \times 1.75)$ and then $\pi \times 1.75 \times (2.1 + 1.05) = \pi(R + r)l$. Do not rush this algebra: allow 15 seconds wait before asking, 'Why does the answer match?' 6) Close with a claim-evidence-reasoning prompt and invite 2 volunteers to share, using the sentence starter 'The artisan needs ... because ...'. 7) Link back: 'Mzee Kiplagat's drum has a curved wall too. What do you think we need to know to find its sheet?' (Do not answer fully; Lesson 4 completes the drum's SA check.)	makes the role of each measurement visible. Learners explain why slant height rather than vertical height belongs in the surface formulae, tying the formulae to what the sheet metal actually covers, which connects directly to the phenomenon's central point that surface area measures material.	learner uses 1.75 m, not 1.4 m or 3.5 m, as the frustum's slant height; (d) learner shows both methods giving 17.325 m²; (e) CER response names the formula, the measurements used, and the reason. Exit-check mini-question on a slip: 'A conical top has $r = 2.1$ m and $l = 3.5$ m. Curved SA?' Expected: $22/7 \times 2.1 \times 3.5 = 23.1$ m². Sort slips into 'correct', 'used h instead of l', 'forgot π or unit' piles to plan the next lesson's starter.
Driving Question Board (DQB) Creation VIDEO: Volumes of cones	(About 6 minutes.) Learners return to the class Driving Question Board, which displays the driving question: 'Why does the same amount of metal sheet make containers that hold very different	1) Point to the DQB and say: 'Today we added tools for curved surfaces. What new questions do they raise for Mzee Kiplagat's puzzle?' 2) Give 2 minutes of silent writing: 'One question per note, in	Driving Question Board as a public record of growing understanding: sorting notes into 'material' and 'space' categories reinforces that surface area and volume are different quantities, and ticking off	Observable indicators: (a) at least one question per group is a genuine new question, not a repeat; (b) learners correctly sort notes into surface-area and volume categories (target: at least

intuition <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Surface area using nets <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	amounts of grain — and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction? Groups write new questions and findings on sticky notes or strips. Examples: 'We can now find sheet for round and tapering shapes: what about a sphere?'; 'The cone and frustum need curved SA; does the drum's wall use a similar idea?'; 'The conical top used 8.8 m² of sheet, but how much grain would the tank hold?'; 'Does the cone with the same sheet as a cylinder hold more or less?' Learners place notes on the board under categories such as 'Surface area tools we have', 'Questions about capacity', and 'Questions about best shape'. They also mark which earlier questions have been partly answered (for example, how to cover a curved surface).	a full sentence.' 3) WAIT TIME 10-15 seconds, then cold-call 4 groups to read one note aloud and say where it belongs. 4) Probe: 'Is that a question about material or about space inside? Which do we measure with surface area?' Cold-call 2 learners. 5) Sort aloud with the class: place notes on 'surface area' or 'volume/capacity' sides. Highlight that the conical tank top's 8.8 m² tells us nothing about how much water the tank holds, so the question 'how much grain will it hold?' stays open for Lessons 5 and 6. 6) Mark one earlier note, such as 'How do we find sheet for a curved surface?', with a tick and today's date, and write 'cone: πrl; frustum: $\pi(R + r)l$' beside it. 7) Remind learners that the volume questions will not be answered today, to keep the board honest.	answered questions shows learners their own progress while keeping unanswered questions (capacity, best shape) as motivation for later lessons.	80% of notes sorted correctly after the teacher probe); (c) learners can name one question that today's work cannot yet answer and say which quantity it needs (volume). Teacher records the number of notes in each category and flags groups that confuse sheet needed with grain held, to address in the next lesson.
Model Building Phase VIDEO: Volumes of cones intuition <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Surface Area and Nets (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	(About 6 minutes.) Learners add to their evolving class and individual models of why the drum and the box differ. Each learner updates a model page with three parts: (1) A diagram of the cone net (sector with l as radius and arc $2\pi r$) and a frustum labelled R, r and l. (2) A short 'what we know' list: Lesson 1: the box's SA is 30,800 cm² from the sum of faces; Lesson 2: slant height and pyramid faces; Lesson 3: curved SA of a cone = πrl and of a frustum = $\pi(R + r)l$, with data 8.8 m² and 17.325 m². (3) A revised 'what we still need' statement, such as 'We still need the drum's SA check (curved wall) and the volumes of all solids to explain why equal sheet gives different grain capacity.' Learners also write one sentence on how curved-surface formulae help an	1) Say: 'Take out your model page. Add what we learned today in your own words, with a diagram.' 2) Display a sample model with blanks and let learners fill in details. Give 3 minutes of silent work. 3) Circulate and look for diagrams where the sector's straight edge is labelled l. Ask: 'Why is the radius of the sector not r?' WAIT TIME 10 seconds; cold-call 2 learners. 4) Ask: 'Does anything on your model yet explain why the drum holds more grain than the box?' Wait 15 seconds; cold-call 3. Affirm answers such as 'No, not yet: so far we only have surface area tools; we need volume.' 5) Ask a stretch question for early finishers: 'If a cone and a cylinder have the same radius, which one needs more sheet for its curved surface, if l is bigger than	Cumulative model revision: learners add the cone and frustum to their growing model, explicitly separating 'what we know' (surface area tools) from 'what we still need' (volume). This keeps the model honest, shows how each lesson adds evidence, and keeps the distinction between material and capacity central to the driving question.	Observable indicators: (a) the model page has a labelled sector with the radius marked as l and arc as $2\pi r$; (b) the frustum is labelled with R, r and slant height l and the learner writes both methods or the formula; (c) the 'what we still need' statement correctly names volume (not more surface area) as the missing piece for the drum vs box puzzle; (d) learners cite the correct Lesson 3 values (132 cm², 8.8 m², 17.325 m²). Scan pages and sort into 'secure', 'needs one reminder', and 'needs re-teaching' groups; plan a 3-minute starter next lesson for the last two groups.

	artisan: 'They tell Wanjiru how much sheet to buy and cut for round shapes.' Learners then complete the summary table prompts (see below) in their notebooks.	h?' and let them explore with numbers. 6) Close with one sentence: 'Wanjiru can now find sheet for flat faces, slanted faces, cones and frustums. Next we do spheres, composite shapes, and the drum's wall.' (Do not teach the sphere formula today.) 7) Collect model pages for a quick scan; note learners who still use h in the cone formula.		
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D. TEACHER REFLECTION

After the lesson, ask yourself: (1) Did learners discover that the cone's net is a sector, or did I tell them too early? Which questions got them to see that the sector's radius is l ? (2) How many learners confused l with h in the formulae, and which prompt (the slant-edge pointing in the Predict phase, or the cross-section drawing in the Explain phase) helped most? (3) Did learners keep curved SA and total SA distinct, and did anyone ask about the base without prompting? (4) Were the frustum checks ($23.1 - 5.775$ and $\pi(R + r)l$) understood as the same idea, or only copied? (5) Did the DQB sorting into 'material' and 'space' show that learners can separate surface area from volume, which is the core idea of the phenomenon? (6) Which learners need extra practice with units (cm^2 vs m^2) and with Pythagoras for l before Lesson 4 (spheres and composite solids)? Note practical issues such as scissors, string supply, and time, and adjust the cutting task (for example, pre-cut cones) if time ran short.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the curved surface of a paper cone, when cut along its slant edge and flattened, forms a sector of a circle. The straight edge of the sector equals the slant height l , and the arc length equals the circumference of the cone's base ($2\pi r$). We also observed that a frustum is a big cone with a smaller cone removed from the top, using the ice-cream cone ($r = 3.5$ cm, $l = 12$ cm), the conical tank top ($r = 1.4$ m, $l = 2$ m) and the Turkana frustum tank.
What did I learn?	We learned that the curved surface area of a cone = πrl , and the total surface area = $\pi r(r + l)$. The frustum's curved surface area = $\pi(R + r)l$, which equals the big cone minus the removed cone. We learned that the ice-cream cone has a curved SA of 132 cm^2 , the conical tank top needs 8.8 m^2 of sheet, and the Turkana frustum needs 17.325 m^2 (checked by $23.1 - 5.775$). We also learned to state whether an area is curved or total and to use slant height l , not vertical height h , in these formulae.
How does this explain the phenomenon?	We can explain that an artisan like Wanjiru finds the sheet needed for cones and frustums by unrolling them into sector nets and using πrl or $\pi(R + r)l$, so surface area tells us how much material covers a solid. This only measures the material, not the space inside, so we still need volume to explain why Mzee Kiplagat's drum and box, made from the same $30,800 \text{ cm}^2$ of sheet, hold different amounts of maize.

LESSON 4 (40 minutes (one double lesson of 80 minutes is recommended if learners make the paper-strip model)): Surface Area of Spheres and Composite Solids: Wrapping the Football, Decomposing the Container

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners find the surface area of a sphere by wrapping a ball in paper strips and then verify with $SA = 4\pi r^2$. They then find the surface area of composite solids (cylinder + hemisphere, as in the Nairobi packaging factory container) by decomposing the solid into parts and removing hidden joined faces, and check the method on the drum (30,800 cm ²). This moves the evidence on from 'SA is the sheet used' toward 'a sphere covers the most space for its skin'.
Knowledge	By the end of the lesson the learner should be able to: (a) state and use sphere $SA = 4\pi r^2$ and hemisphere curved $SA = 2\pi r^2$; (b) determine the surface area of composite solids by decomposing them into familiar solids; (c) identify and leave out the faces hidden where solids are joined; (d) state clearly whether total or curved surface area is being found. (KICD ability: determine the surface area of prisms, pyramids, cones, frustums and spheres, and composite solids.)
Skills	Measuring (circumference, diameter and radius of a football), modelling with paper strips, decomposing composite solids, computing with $\pi = 22/7$, justifying a method with evidence (NGSS SEP: Developing and Using Models; Analyzing and Interpreting Data; Using Mathematics and Computational Thinking; Constructing Explanations).
Attitudes	Responsibility: the learner takes care of the footballs and models used. Integrity: the learner measures and records accurately. Unity: the learner works with a team and respects each member's input. Appreciation of mathematics used by Kenyan fabricators and packaging engineers.
Key Inquiry Question	1. How do we determine the surface area and volume of solids? 2. How are surface area and volume of solids applied in real-life situations?
Purpose in Storyline	Lessons 1-3 built the sum-of-faces and curved-area methods (prisms, pyramids, cones, frustums). Lesson 4 adds the sphere and composite solids. Learners check the drum SA (30,800 cm ²) with a decomposition, which confirms that surface area is only the material, and it sets up Lessons 5-6, where volume explains why the drum holds more. The sphere also supports the later claim that a sphere is the best shape of all.
Safety Notes	Keep footballs, rulers and scissors off the floor so no one trips. Scissors are used for paper strips only, with the pointed end down. Use water-soluble markers. Do not throw or kick the balls in the classroom. Learners with limited mobility can take the role of recorder or checker.

B. LESSON OVERVIEW

Mzee Kiplagat is still puzzled: the drum and the box both used 30,800 cm² of Wanjiru's iron sheet, yet the drum holds more maize. Today learners turn to a shape with no flat faces, a football from a Kisumu school ($r = 10.5$ cm). They predict how many paper circles of the same radius are needed to cover it, wrap it in strips to find that the answer is about 4, and confirm $SA = 4\pi r^2 = 1,386$ cm². They next meet the Nairobi packaging factory container (cylinder $r = 7$ cm, $h = 10$ cm with a hemisphere on top, no lid between), decompose it, and find that the joined circular face is hidden and must not be counted: $SA = 440 + 154 + 308 = 902$ cm². Finally they re-decompose the drum ($2\pi r^2 + 2\pi rh = 7,700 + 23,100 = 30,800$ cm²) to check that the method reproduces the anchor figure, and they add new evidence to the DQB. Lesson 4 teaches only surface area; volume of the cylinder, drum and sphere is left to Lessons 5 and 6.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Composite Solids Principles - Basic Source: CK-12  Search ARES for similar videos  HTML: Surface Area of Spheres (Flexbook) Source: CK-12  Search ARES for similar readings	(8 min) Learners meet Wanjiru's problem again: 'A packaging factory in Nairobi wants to cover a football ($r = 10.5$ cm) in a thin film.' Each learner holds a football (or a ball of known radius) beside a paper circle cut with the same radius. Individually they write a prediction in their books: 'How many paper circles of radius 10.5 cm would I need to cover the whole football with no overlap?' with a reason, then sketch the covering. They guess on a sticky note (2, 3, 4, 5 or 6) and add it to a class tally on the board. They also predict: 'If I join a hemisphere on top of a cylinder, should I count the circle where they meet in the surface area?'	Hold up the football and say: 'Last time we found the material on a cone and a frustum. A football has no flat face and no net that lies flat. How could Wanjiru find the sheet needed for it?' Give the instruction: 'Write your own prediction first. No talking for 2 minutes.' WAIT TIME 15 seconds after asking: 'Why might you need more than one circle?' Cold-call 4 learners to give their number and reason, including at least 1 who chose a low number and 1 who chose a high number. Do not say which is right. Record the tally on the board as a bar graph. Then ask: 'Is a hidden face part of the sheet metal? Raise a hand for yes, no, or not sure.' Note counts for use in the Explain Phase. Link back: 'Remember the drum and box: equal sheet, unequal grain. Keep that in mind.'	Predict-Observe-Explain (Prediction with Reasoning): committing in writing to a prediction and reason before seeing evidence exposes the intuition that a sphere's skin is 'about 2 or 3 circles', so the later evidence creates a real conflict. The class tally turns private guesses into shared data that learners later test against measurements.	Look for: (1) every learner has a written prediction with a reason before the tally; (2) at least 3 learners mention comparing the curved ball to flat paper circles; (3) the count of learners who say the joined face is or is not part of the sheet. Note learners who say 'surface area is the space inside' and revisit the SA vs volume distinction with them. Record the tally as a baseline.
Observe Phase  VIDEO: Composite Solids Principles - Basic Source: CK-12  Search ARES for similar videos  EXERCISE: Surface Area of Spheres Practice Source: CK-12  Search ARES for similar readings	(14 min) Groups of 4 collect evidence in two stations. Station A (sphere): (1) Measure the football circumference with string and a ruler (about 66 cm) and compute the radius using $C = 2\pi r$: $r = 66 \div (2 \times 22/7) = 10.5$ cm. (2) Cover half the ball with strips of paper or masking-tape strips of known width, laying them side by side without overlap, then peel them off and lay them flat. (3) Cut the strips to fit into circles of radius 10.5 cm (area $\pi r^2 = 346.5$ cm ²) and count how many circles' worth are used (about 4). Record: strip area $\div$ circle area. (4) Compute $4\pi r^2 = 4 \times 22/7 \times 10.5 \times 10.5 = 1,386$ cm ² and compare with $4 \times 346.5 = 1,386$ cm ² .	Circulate with a checklist. Say to each group: 'Show me where you laid the strips so they do not overlap. How do you know there is no gap?' At Station A ask: 'About how many circle-areas did your strips take? Write it down before you compute anything.' WAIT TIME 10 seconds after: 'What do you notice about your number compared with our tally on the board?' At Station B ask: 'Point to the face that is covered when the hemisphere sits on the cylinder. Can the metalsmith see it? Does Wanjiru need sheet for it?' Prompt: 'Which areas did you add? Which did you leave out? Why?' Remind: 'Always say whether you are finding the total or the curved	Gathering Evidence Through Physical Modelling and Data Pooling: wrapping the ball makes a curved surface measurable, so learners see 4 circle-areas as a measurement rather than a rule. Colour-coding and the hidden-face label make the decomposition concrete, building the idea that composite surface area is the sum of the visible parts only. Pooling group ratios lets learners judge measurement error and agreement.	Observable indicators: (1) strips laid with no overlap or gap; (2) the group's ratio is close to 4 (accept 3.8 to 4.2) and learners say why small differences arise; (3) the hemisphere is coloured separately and the joined circle is marked hidden before any calculation; (4) the table shows 440, 154 and 308 with units cm ² ; (5) learners say 'curved' or 'total' when naming areas. Use a tick sheet. If more than 1 group counts the joined circle, plan a short re-model in the Explain Phase.

	Station B (composite solid): Using a model of the Nairobi packaging container made from a rolled-up card cylinder ($r = 7$ cm, $h = 10$ cm) with a card hemisphere on top (or a drawn diagram with the dimensions), learners label the parts: curved cylinder surface, base of the cylinder, and the hemisphere. They colour-code each part. They place the hemisphere on the cylinder and identify the circle where they meet ($r = 7$ cm) and mark it 'hidden'. They compute each part: curved cylinder $2\pi rh = 2 \times 22/7 \times 7 \times 10 = 440$ cm²; base $\pi r^2 = 154$ cm²; hemisphere curved $2\pi r^2 = 308$ cm². They record these in a table.	surface area.' Cold-call 3 groups to report their strip-to-circle ratio and check that they cluster near 4. For a group that gets 3 or 5, ask them to check overlaps or gaps. Rotate stations after 7 minutes. For learners who struggle, offer a pre-labelled diagram and the formula card from Lesson 3.		
Explain Phase  VIDEO: Cylinder volume & surface area <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Surface Area of Spheres (Flexbook) Source: CK-12  Search ARES for similar readings	(12 min) Part 1 (4 min): Using the pooled data, each group writes a claim: 'The surface area of a sphere is ____ times the area of its great circle, so $SA = ___$.' They back it with the strip evidence and the numbers 346.5 cm² and 1,386 cm², and write $4\pi r^2$. Part 2 (4 min): Groups complete the Nairobi container: $440 + 154 + 308 = 902$ cm² and write why the circle where the solids join (154 cm², top of the cylinder, and the flat face of the hemisphere) is left out. They explain why the container has no lid between the cylinder and the dome. Part 3 (4 min): Groups check the method on the drum ($r = 35$ cm, $h = 105$ cm, closed): $2\pi r^2 = 7,700$ cm² for the two circular faces, $2\pi rh = 23,100$ cm² for the curved wall, TSA = 30,800 cm². They compare this with the box's 30,800 cm² from Lesson 1 and write: 'Both containers use the same sheet, so surface area does not tell us capacity.'	Say: 'Use your own data. Start your claim with: We found that...' Give each group 2 minutes of silent writing before sharing. WAIT TIME 15 seconds after asking: 'Why does $4\pi r^2$ make sense when the strips took about 4 circles?' Cold-call 3 groups for the sphere claim, then 2 for the hidden-face reasoning: 'Why did the Nairobi container lose 154 cm² at the join, in other words why is it not $440 + 154 + 154 + 308$?' Ask for a revoicing: 'Who can say in other words what Group 2 said?' Contrast carefully: 'Hemisphere curved SA is $2\pi r^2$, half of the sphere; check that against $308 = \text{half of } 616$.' For the drum say: 'Which parts did we add? Did we include any hidden faces? Do we get the same 30,800 as Wanjiru's sheet?' Address any misconception recorded earlier: 'Is surface area the space inside?' Use the drum and box to reinforce: 'Same sheet, different grain, so what must the sheet measure?	Claim-Evidence-Reasoning (CER) with Revoicing: learners must link the pooled strip measurements (evidence) to the formula (claim) and explain why it works (reasoning). Revoicing and cold-calling spread ideas across the class. Returning to the drum SA is a Transfer Check: it shows the decomposition method reproduces a known anchor value, building trust in the method.	Look for: (1) claims that use measured data (e.g. 'about $4 \times 346.5 = 1,386$') not just the formula; (2) a correct explanation that the joined circle is hidden and not part of the outside sheet; (3) correct results 1,386, 902 and 30,800 cm² with units; (4) learners use 'total' or 'curved' correctly; (5) at least 4 learners can explain why equal SA does not mean equal capacity. Collect a quick exit question: 'A hemisphere of $r = 7$ cm sits on a cylinder of $r = 7$ cm. Which areas do you add?' (Expected: curved cylinder + one base + curved hemisphere.) Revisit if fewer than 80% are correct.

		What might measure the space inside? We will come to that in Lessons 5 and 6.' Do not teach volume formulae today.		
Driving Question Board (DQB) Creation  VIDEO: Volume of a sphere <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Surface Area of Spheres Practice Source: CK-12  Search ARES for similar readings	(3 min) Groups revisit the DQB. Each group writes on sticky notes: (1) one thing they now know, e.g. 'A sphere's SA is 4 great circles: 1,386 cm² for the football' or 'Hidden joined faces are not counted: 902 cm²'; and (2) one new question, e.g. 'The football and the drum have different shapes. Which holds more for the same skin?' or 'Why would a packaging engineer put a dome on a container?' or 'What is the volume of the drum, and how do we find it?' They place the notes next to older ones and move any answered question to an 'Answered' column, such as 'How do we find SA of a curved shape?'	Say: 'Our board started with Mzee Kiplagat's puzzle. What can we now add?' Ask groups to say whether their question is about surface area (material) or about volume (space inside) and place it in the matching column. WAIT TIME 10 seconds before collecting. Cold-call 3 groups to read a note aloud. Point out: 'Which questions can only be answered when we know the space inside? Lesson 5 starts with the box. Lesson 6 gives us the drum.' Highlight good questions and group like questions together. Photograph the DQB for the class record.	DQB Sorting and Tracking: separating 'surface area' questions from 'volume' questions on the board reinforces that they are different quantities. Moving answered questions shows progress, and new questions give the next lessons a purpose.	Indicators: each group posts at least 1 'now know' note and 1 new question; at least 2 questions correctly sorted as volume questions; at least 1 question links shape to capacity (e.g. 'which shape holds the most?'). If notes only restate formulae, prompt with 'How does this help Mzee Kiplagat?'
Model Building Phase  VIDEO: Composite Solids Principles - Basic Source: CK-12  Search ARES for similar videos  EXERCISE: Volume and surface area of cylinders <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(3 min) Learners update their individual class model (a flow diagram or concept map, started in Lesson 1) with a new panel titled 'Surface area is the sheet used'. They add: sphere SA = $4\pi r^2$ (football 1,386 cm²); hemisphere curved SA = $2\pi r^2$; composite solids: decompose, add the visible parts only, hidden joined faces removed (Nairobi container 902 cm²); drum TSA = 30,800 cm² = box TSA. They draw an arrow labelled 'But what about the space inside? Lessons 5 and 6' pointing to a blank 'Volume' panel. They also write a 2-sentence answer to: 'What have we learned that explains why Mzee Kiplagat's drum and box cost the same sheet?'	Say: 'Add today's evidence to your model. Draw the football and the Nairobi container so we remember the hidden face.' Circulate and ask: 'Where on your model does the sheet used show up? Where will the grain show up?' WAIT TIME 10 seconds. Cold-call 2 learners to read their 2-sentence explanation. Reinforce: 'So far our model explains the material. It cannot yet explain why the drum holds more. What do we still need?' Tell learners to bring a cylinder or box from home (e.g. a tin of beans, a milk carton) for Lesson 5, and to be ready to fill models with sand or maize.	Cumulative Model Revision: learners add each lesson's evidence to one evolving model. Showing a deliberate gap (the blank volume panel) makes the limits of current understanding visible and gives the next lessons a clear purpose.	Check the models for: correct formulae and values (1,386 cm², 902 cm², 30,800 cm²); the hidden face noted; the arrow to volume present; a written explanation that does not mix up surface area and volume. Collect 5 models at random to spot misconceptions, and give feedback at the start of Lesson 5.

D. TEACHER REFLECTION

After the lesson ask yourself: (1) Did learners get about 4 for the strip-to-circle ratio, and could they explain differences in terms of gaps or overlaps? (2) How many groups wrongly counted the joined circle in the Nairobi container, and did the colour-coding help? (3) Did anyone use 'surface area' and 'volume' interchangeably, and what did I do about it? (4) Did cold-calling bring out learners who usually stay quiet? (5) Were the questions on the DQB moving toward volume and shape? (6) What would I change next time, for example more time at Station B, or a pre-cut hemisphere to save time? Note names of learners who need extra support with decomposing solids and plan quick practice using the Lesson 3 formulae before Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We wrapped a football ($r = 10.5$ cm) in strips and found the skin covered about 4 circles of radius 10.5 cm. We saw that the Nairobi container is a cylinder with a hemisphere on top, with a hidden joined circle. We added up the parts of the drum and got 30,800 cm ² , the same as the box.
What did I learn?	Sphere $SA = 4\pi r^2$ (football 1,386 cm ²). Hemisphere curved $SA = 2\pi r^2$. For composite solids we decompose into familiar parts, add only the outside faces, and leave out the hidden joined faces (Nairobi container 902 cm ²). Surface area is the amount of material covering a solid, not the space inside.
How does this explain the phenomenon?	The drum and the box both use 30,800 cm ² of sheet, so surface area alone cannot tell Mzee Kiplagat which holds more maize. Surface area measures the material covering a solid, so we still need to find the space inside (volume) in Lessons 5 and 6 to explain why the drum holds more.

LESSON 5 (40 minutes (1 double lesson or 2 x 40 minutes)): Volume of Prisms and Pyramids: How Much Maize Can the Box Hold?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners return to Mzee Kiplagat's two containers. Lessons 1 to 4 dealt with surface area (the material covering a solid). In this lesson learners turn to volume (the space inside). They derive $V = \text{base area} \times \text{height}$ for prisms and $V = \frac{1}{3} \times \text{base area} \times \text{height}$ for pyramids from sand-fill models, then compute the box volume of 322,500 cm ³ and preview the drum.
Knowledge	By the end of the lesson the learner should be able to: (a) determine the volume of prisms and pyramids; (b) explain that volume (space inside, in cm ³ or m ³) is different from surface area (material covering, in cm ² or m ²); (c) state that three pyramids fill a prism with the same base and height; (d) convert cm ³ to litres (1 litre = 1,000 cm ³ ; 1 m ³ = 1,000 litres).
Skills	Measuring dimensions; comparing filled models; deriving a formula from evidence; calculating volume with correct units; converting between cm ³ , m ³ and litres; arguing from evidence. NGSS-aligned practices: Planning and Carrying Out Investigations; Analyzing and Interpreting Data; Using Mathematics and Computational Thinking; Constructing Explanations.
Attitudes	Values: Responsibility (care for sand, water and models, and sweeping up spills) and Integrity (recording measurements and fill counts honestly). Core competencies: Critical Thinking and Problem Solving (reasoning from fill evidence to a formula); Communication and Collaboration (sharing roles and results in groups). PCI: Education for Sustainable Development (reusing sand and water and not wasting materials).
Key Inquiry Question	1. How do we determine the surface area and volume of solids? 2. How are surface area and volume of solids applied in real-life situations?
Purpose in Storyline	Lessons 1 to 4 built the surface area evidence (box SA = 30,800 cm ² , equal to the drum). Lesson 5 starts the volume evidence: learners derive the prism and pyramid rules and calculate the box volume of 322,500 cm ³ (322.5 litres). Lesson 6 then uses the one-third idea with cones and spheres to find the drum volume (404,250 cm ³) and resolve why the drum holds more.
Safety Notes	Use dry sand or water only over trays or basins. Wipe up spills immediately because wet floors cause slips. Do not throw sand. Keep sand away from eyes and wash hands afterwards. Handle card or plastic models gently and do not use cut edges carelessly. No learner should lift a full container; use small models only.

B. LESSON OVERVIEW

Teacher recalls Mzee Kiplagat and Wanjiru: both containers use 30,800 cm² of sheet, yet the drum holds more maize. Learners predict whether the box or the drum holds more and what a pyramid holds compared with a prism. In the Observe phase they fill the 5 x 4 x 3 cm and 3-4-5 triangular prism models (both 60 cm³) with sand or water and compare them. They then fill a pyramid that has the same base and height as a prism and count how many pyramid-fuls fill the prism (3). In the Explain phase they calculate the box volume ($50 \times 50 \times 129 = 322,500 \text{ cm}^3 = 322.5 \text{ litres}$) and the pyramid roof volume (48 m³), and distinguish volume from surface area. The DQB and the class model are then updated with volume evidence, and the drum volume is left for Lesson 6.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Volume of pyramids intuition Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Volume of prisms and pyramids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners see the photograph of Mzee Kiplagat's drum and box in Eldoret. Individually (2 minutes) they write: (1) Which container will hold more maize, the box (50 cm x 50 cm x 129 cm) or the drum? Why? (2) Wanjiru says the box has a surface area of 30,800 cm ² , so does that tell us how much maize it holds? (3) A pyramid and a prism have the same square base and the same vertical height. Predict how many pyramid-fuls of sand will fill the prism. They sketch the prediction in their notebooks. They share in pairs.	Display the photograph and say: "Mzee Kiplagat is still puzzled. Wanjiru used exactly 30,800 cm ² of sheet for each container. Today we find out what is inside, not what is outside." Ask: "Is surface area the same as the space inside?" Give WAIT TIME of 10-15 seconds before taking answers. Cold-call 4 learners for their prediction on the pyramid-fuls and record every answer on the board (e.g. 1, 2, 3, 4, 'half') without judging. Say: "Keep your prediction; the sand will tell us." Circulate and note learners who think equal sheet means equal volume.	Predict-Observe-Explain (prediction commitment): writing a prediction first exposes the misconception that equal surface area means equal volume and the guess about pyramids, so that learners later have to compare the sand evidence with their own ideas.	Observable indicators: Does the learner write a prediction with a reason rather than leaving it blank? Does the learner use words like 'inside', 'space' or 'capacity', or still say 'size' for both surface area and volume? Tally on a class sheet how many predicted 3 pyramid-fuls. Learners who write cm ² for amount of maize need a reminder in the Explain phase.
Observe Phase  VIDEO: Measuring volume as area times length Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Surface Area and Volume of Pyramids Practice Source: CK-12  Search ARES for similar readings	Groups of 4 with roles (measurer, filler, recorder, reporter). Station A: measure the 5 cm x 4 cm x 3 cm rectangular prism model and the triangular prism with 3-4-5 cm right-angled triangle ends and length 10 cm. Record base area and height or length (rectangle 5 x 4 = 20 cm ² base, height 3 cm; triangle 1/2 x 3 x 4 = 6 cm ² base, length 10 cm). Fill each model with dry sand (or water) from a measuring cylinder and record the volume: both hold 60 cm ³ . They compare: 'different shapes, same volume'. Station B: take an open pyramid and an open prism with the same square base and the same vertical height. Fill the pyramid and pour it into the prism, repeating and counting the number of pyramid-fuls until the prism is full. Record: 3 pyramid-fuls fill the prism. Station C: the recorder completes a table of shape, base area, height, base area x height, sand volume (and 1/3 of that for the pyramid). Groups look for	Distribute materials and say: "Measure first, then fill. Level the sand at the rim; do not heap it." Circulate with the prompts: "Which length is the height of the prism? Which is the base?" and "Is the triangular prism's volume the same as the rectangular one? How did you check?" Give WAIT TIME of 10-15 seconds after each question before helping. Remind groups to use the vertical height of the pyramid, not the slant height (as in Lesson 2). Stop one group that gets 2.5 or 3.5 pyramid-fuls and ask: "Was the sand levelled? Did you repeat it?" Cold-call 3 groups to report their number of pyramid-fuls and write all results on the board. Ask: "What do you notice about base area x height for the two 60 cm ³ prisms?"	Structured data collection and pattern finding (investigation with a results table): learners generate their own quantitative evidence and look for a repeated pattern (base area x height = sand volume; 3 pyramids = 1 prism), which builds the formulae from evidence instead of being given them.	Observable indicators: Learners measure the base and height correctly and compute the base area (20 cm ² and 6 cm ²). Learners level the sand and repeat the fill count. The recorder's table shows 60 cm ³ for both prisms and 3 pyramid-fuls. Use a checklist: correct base area (Y/N), correct height used (Y/N), pyramid count recorded (Y/N). Learners who use the slant height or the sloping edge for the pyramid are asked to find the vertical height.

	patterns in the table.			
Explain Phase  VIDEO: Volume of a rectangular prism Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Surface Area and Volume of Pyramids (PLIX) Source: CK-12  Search ARES for similar readings	Whole class then pairs. Learners use the table to state the rules in their own words: prism volume = base area x height ($V = Ah$); pyramid volume = $\frac{1}{3} \times$ base area x height because 3 pyramids fill the prism. Apply: (1) Box from Mzee Kiplagat: base $50 \times 50 = 2,500 \text{ cm}^2$; height 129 cm; $V = 2,500 \times 129 = 322,500 \text{ cm}^3 = 322.5$ litres. (2) Pyramid roof from Lesson 2: base $6 \text{ m} \times 6 \text{ m} = 36 \text{ m}^2$; vertical height 4 m; $V = \frac{1}{3} \times 36 \times 4 = 48 \text{ m}^3 = 48,000$ litres of air space under the roof. They check that they used the vertical height 4 m and not the slant height 5 m. (3) Convert 322,500 cm^3 to litres by dividing by 1,000. (4) Discuss in pairs: the box SA is 30,800 cm^2 and its volume is 322,500 cm^3. Why can these numbers not be compared? Learners write a one-sentence difference: surface area is the material covering the solid (cm^2); volume is the space inside it (cm^3). Preview: the drum is a cylinder, which is like a prism with a circular base, so $V =$ base area x height should also work. Learners write down a prediction for the drum with $r = 35 \text{ cm}$ and $h = 105 \text{ cm}$, but do not calculate it yet because that is Lesson 6.	Ask: "Using your table, what rule connects the base, the height and the sand volume?" WAIT TIME of 10-15 seconds, then cold-call 3 learners to state the rule and 2 to explain why the pyramid has $\frac{1}{3}$. Model the box calculation on the board, asking learners to supply each step. Ask: "Why is the unit cm^3 and not cm^2?" Ask: "A learner says the pyramid roof volume is $\frac{1}{3} \times 36 \times 5$. What is wrong?" (5 m is the slant height, not the vertical height.) Cold-call 2 learners to explain. Say: "Mzee Kiplagat can now tell you how many litres of maize the box holds: 322.5." Ask: "If the box and drum have the same sheet, can the surface area tell us the volume?" Give WAIT TIME and cold-call 3 pairs. Remind learners that the drum volume is for the next lesson. Do not give the drum answer.	Claim-Evidence-Reasoning with unit sense-making: learners state the rule (claim), cite the sand-fill counts and table (evidence), and justify why $\frac{1}{3}$ and why volume units are cubic (reasoning). This builds the distinction between surface area and volume, which is central to the phenomenon.	Observable indicators: Learners calculate $2,500 \times 129 = 322,500 \text{ cm}^3$ and convert to 322.5 litres correctly. Learners use $h = 4 \text{ m}$ (not 5 m) for the pyramid roof and obtain 48 m^3. Learners write cm^3 or m^3 for volume and cm^2 for area. Exit question: 'A prism has base area 40 cm^2 and height 9 cm; a pyramid has the same base and height. State both volumes.' (360 cm^3 and 120 cm^3.) Mark each as secure, partly secure or not yet.
Driving Question Board (DQB) Creation  VIDEO: Volume of pyramids intuition Source: Khan Academy (English - US curriculum)  Search ARES	Learners add to the class Driving Question Board. On sticky notes they write (a) a question they can now answer (e.g. 'How much maize does the box hold?') and move it to the 'answered' column with the answer 322.5 litres; (b) a new question (e.g. 'Does base area x height work for a round base like the drum?', 'Why does a	Say: "Go to the DQB. Which of our questions did the sand help us answer today?" Give WAIT TIME of 10-15 seconds. Cold-call 4 learners to place a note and say why. Point to the question 'Why does the drum hold more than the box if the sheet is the same?' and say: "We know the box now. What do we still need for the drum?"	Driving Question Board tracking: learners sort questions into answered and open, which shows what the evidence has explained and what is still needed to resolve the phenomenon, and keeps them motivated for Lesson 6.	Observable indicators: New questions refer to the drum, circles or shape (not just 'what is volume?'). Learners link their note to evidence (sand, table or calculation). Count how many learners moved at least one note to 'answered' and how many posted a new question about the drum or about equal sheet with

for similar videos EXERCISE: Surface Area and Volume of Prisms Practice Source: CK-12 Search ARES for similar readings	pyramid hold only a third?', 'Which shape holds the most for the same sheet?'). They group their questions under headings such as 'Surface area', 'Volume' and 'Shape'.	Highlight the link: the drum has a circular base, and the cone-to-cylinder one-third idea is coming in Lesson 6. Add a column 'Evidence from Lesson 5' and record 'Prism $V = \text{base area} \times \text{height}$ ' and 'Pyramid $V = \frac{1}{3} \text{ base area} \times \text{height}$ '.		different volume.
Model Building Phase VIDEO: Measuring volume as area times length Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Volume as area of base times height Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners revise the class model of the phenomenon in their notebooks or on a chart. The model has two columns, 'Surface area (material, cm ²)' and 'Volume (space inside, cm ³)', with the drum and box in each. Learners now fill in the box: SA 30,800 cm ² and V 322,500 cm ³ (322.5 litres). For the drum they write SA 30,800 cm ² (equal) and V '?' with the prediction from the Explain phase. They add arrows and labels: 'prism $V = Ah$ ', 'pyramid $V = \frac{1}{3} Ah$ ', 'same SA does not mean same V'. In pairs they write a three-sentence explanation for Mzee Kiplagat of what is known now and what is still missing.	Say: "Update your model. What can you now add for the box, and what is still a question mark for the drum?" Give WAIT TIME of 10-15 seconds. Ask: "Where in your model does the sand evidence appear?" Cold-call 3 learners to share their explanation for Mzee Kiplagat and ask the class to add one idea from each. Correct any model that mixes up cm ² and cm ³ . Close with: "Next lesson we use cones and spheres, and a third idea, to find the drum volume and see which container holds more."	Cumulative model building: learners add the volume evidence to the model begun in Lesson 1, separating surface area from volume, so the model grows toward the final explanation of why equal sheet gives unequal grain capacity.	Observable indicators: The model shows box $V = 322,500 \text{ cm}^3$ (322.5 litres) and SA 30,800 cm ² under the correct headings with correct units. The drum volume is left open or marked as a prediction. The explanation says that surface area and volume are different and that shape affects volume. Use a 3-point check: correct numbers, correct units, correct distinction.

D. TEACHER REFLECTION

Did learners level the sand and repeat the pyramid fill, so that the 3 pyramid-fuls result was convincing? Which learners still confused vertical height with slant height or cm² with cm³? Did any learners say that the prism and pyramid volumes were equal because of the same base? What evidence showed that learners understood why the drum volume cannot be found from its surface area? What needs to be revisited at the start of Lesson 6 (circle area with $\pi = \frac{22}{7}$, the $\frac{1}{3}$ rule with cones) so that the drum calculation is secure?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We filled the 5 x 4 x 3 cm and 3-4-5 triangular prism models with sand and both held 60 cm ³ . A pyramid with the same base and height as a prism had to be filled 3 times to fill the prism. The box from Eldoret measures 50 cm x 50 cm x 129 cm.
What did I learn?	Volume of a prism = base area x height; volume of a pyramid = $\frac{1}{3}$ x base area x height, using the vertical height. Volume is

	measured in cm ³ or m ³ , and 1 litre = 1,000 cm ³ . The box holds 322,500 cm ³ = 322.5 litres, and the pyramid roof encloses 48 m ³ .
How does this explain the phenomenon?	The box and drum use the same 30,800 cm ² of sheet, but surface area only measures the material covering a solid, not the space inside. The box volume of 322,500 cm ³ comes from base area x height, so equal surface area does not give equal volume. We still need the drum volume (Lesson 6) to see which container holds more maize.

LESSON 6 (80 minutes (2 x 40-minute periods)): Volume of Cones, Frustums and Spheres: Resolving Why the Drum Holds More Maize

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use matching cone and cylinder models to establish that a cone holds one third of a cylinder with the same base and height, apply $V = \frac{1}{3}\pi r^2 h$, $V = \frac{4}{3}\pi r^3$ and frustum = big cone minus small cone, and use these ideas to compute the drum volume and settle why the drum holds more maize than the box of equal surface area.
Knowledge	By the end of the lesson the learner should be able to: (a) determine the volume of cones, frustums and spheres (KICD SLO on volume of solids); (b) state that $V(\text{cone}) = \frac{1}{3}\pi r^2 h$, $V(\text{sphere}) = \frac{4}{3}\pi r^3$ and $V(\text{frustum}) = V(\text{big cone}) - V(\text{small cone})$; (c) calculate the drum volume 404,250 cm ³ (404.25 litres) and compare it with the box volume 322,500 cm ³ (322.5 litres) after confirming both SAs are 30,800 cm ² ; (d) distinguish surface area (material covering a solid) from volume (space inside).
Skills	Developing and using models (matching cone and cylinder, water or sand pouring); planning and carrying out investigations; analysing data (616 cm ³ vs 1,848 cm ³); using mathematics and computational thinking ($\pi = 22/7$); constructing explanations; engaging in argument from evidence. Core competencies: Critical Thinking and Problem Solving; Communication and Collaboration; Learning to Learn.
Attitudes	Values: Integrity (reporting measurements and pours honestly, as Wanjiru measured and did not guess); Responsibility (conserving water or sand and tidying the work area); Unity (sharing roles in groups). PCIs: Education for Sustainable Development (no spilled water, reuse of materials); Life Skills (decision making when choosing a container shape).
Key Inquiry Question	1. How do we determine the surface area and volume of solids? 2. How are surface area and volume of solids applied in real-life situations?
Purpose in Storyline	Lessons 1 to 4 showed that the drum and the box each use 30,800 cm ² of iron sheet. Lesson 5 gave the box volume (322,500 cm ³) and the one-third rule for pyramids. Lesson 6 supplies the missing piece: the volume formulas for round solids, so learners can finally calculate the drum volume (404,250 cm ³), compare it with the box and formally resolve the phenomenon. It also prepares Lesson 7 (composite solids) and Lesson 8 (design project).
Safety Notes	Use small plastic or card cone and cylinder models only. Wipe up spilled water immediately to prevent slipping, or use dry sand or maize in trays. Do not throw footballs or models in the classroom. Handle metal sheet offcuts, if used, with care because of sharp edges.

B. LESSON OVERVIEW

Learners begin by recalling the unresolved puzzle: Wanjiru, the jua kali metalsmith in Eldoret, used exactly 30,800 cm² of sheet for both the drum and the box, and the box volume is 322,500 cm³. What is the drum volume? They predict how many cone-fuls fill a cylinder of the same base and height, then pour water or sand with matching models ($r = 7$ cm, $h = 12$ cm) and find that 3 cone-fuls fill the cylinder: 616 cm³ vs 1,848 cm³. They derive $V = \frac{1}{3}\pi r^2 h$, then accept $V = \frac{4}{3}\pi r^3$ for a sphere (football $r = 10.5$ cm gives 4,851 cm³) and build the frustum volume as big cone minus small cone using the Turkana water tank (11.319 m³ = 11,319 litres). Finally they compute the drum volume with $V = \pi r^2 h = 404,250$ cm³, compare it with the box, and explain with evidence that equal surface area does not mean equal volume. They update the DQB and the class model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Volumes of cones intuition Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Volume of cylinders, spheres, and cones word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the phenomenon: Mzee Kiplagat's drum and box in Eldoret, each made by Wanjiru from 30,800 cm ² of iron sheet. They are reminded that the box holds 322,500 cm ³ (Lesson 5). Individually they write: (1) a prediction of whether the drum holds more than, less than or the same as the box, with a reason; (2) how many cone-fuls of sand they think will fill a cylinder with the same radius and height; and (3) a sketch of the two models labelled 'cone' and 'cylinder'. They then share with a partner.	Display the drum ($r = 35$ cm, $h = 105$ cm) and box (50 cm x 50 cm x 129 cm) sketches and say: 'Both used 30,800 cm ² of sheet. Lesson 5 told us the box holds 322,500 cm ³ . Today we find out about the drum.' Pose: 'How many cone-fuls do you think fill this cylinder if they have the same base and height?' Give WAIT TIME of 10-15 seconds of silent writing before any talk. Cold-call 4 learners for predictions, recording each on the board without judging; ask 2 of them 'What makes you think so?'. Hold up the matching cone and cylinder ($r = 7$ cm, $h = 12$ cm) and say: 'Keep your prediction; the models will be our evidence.'	Predict-Observe-Explain (POE) with Think-Pair-Share: committing to written predictions exposes the intuition that equal material means equal capacity, and it gives learners a stake in the later evidence, so they notice when it confirms or contradicts their ideas.	Check that at least 80% of learners have a written prediction and reason before sharing. Look for learners who say 'same, because the same sheet was used' (surface area/volume confusion) and note names for follow-up. Listen for whether any learner mentions 'a third' or 'half' for the cone and record the distribution of cone-ful predictions on the board.
Observe Phase  VIDEO: Volume of a sphere Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Volume of cylinders, spheres, and cones word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4 with roles (pourer, measurer, recorder, reporter), learners use matching hollow cone and cylinder models ($r = 7$ cm, $h = 12$ cm). They first measure and record r and h to confirm both match. They fill the cone to the brim with water (or dry sand) and pour it into the cylinder, counting cone-fuls until the cylinder is full, and repeat to check. They find 3 cone-fuls fill the cylinder. They then calculate cylinder volume = $22/7 \times 7 \times 7 \times 12 = 1,848$ cm ³ and cone volume = 616 cm ³ , and compare. Next, learners study the football from a Kisumu school ($r = 10.5$ cm) and the given result $V = 4/3\pi r^3 = 4/3 \times 22/7 \times 10.5 \times 10.5 \times 10.5 = 4,851$ cm ³ , and check the arithmetic. Finally they look at the Turkana water tank frustum: big cone $r = 2.1$ m, $h = 2.8$ m (volume 12.936 m ³) and removed small cone $r = 1.05$ m, $h = 1.4$ m	Circulate and say: 'Show me that your cone and cylinder have the same radius and the same vertical height, not slant height.' Insist on 'level to the brim' and 'no spills'. Ask each group: 'How many cone-fuls filled the cylinder? Did your second pour agree?' Give WAIT TIME of 10-15 seconds after asking 'Why do you think the answer is a whole number and not 2.5?'. Cold-call 3 groups to report their pour counts and write all results on the board; ask 'What do all groups' results have in common?'. For the sphere, ask: 'The football is 4,851 cm ³ . If we wrapped it in a cylinder of the same radius and height $2r$, what fraction might it fill?' (hint only). For the frustum, ask: 'Which volume do we subtract, and why?' Remind learners to state units (cm ³ , m ³ , litres) and use $\pi = 22/7$. Support slower groups with a prepared	Evidence Table and Consensus Data: groups pool pour counts into a class table (3 cone-fuls each) so the pattern emerges from repeated independent trials, then connect it to the computed 616 cm ³ and 1,848 cm ³ . Using one reasoning chain (big minus small) for the frustum links back to Lesson 3 and avoids a new formula to memorise.	Observe whether groups verify matching r and h before pouring and record 3 cone-fuls. Check that $1,848 / 616 = 3$ is stated unprompted. Check that recorders write units correctly (cm ³ not cm ²) and that the frustum subtraction uses $12.936 - 1.617 = 11.319$ m ³ . Note groups that confuse slant height with vertical height; give a prompt card: 'Volume uses h , not l .'

	(volume 1.617 m ³), and calculate 12.936 - 1.617 = 11.319 m ³ = 11,319 litres.	table: r, h, V(cylinder), V(cone).		
Explain Phase  VIDEO: Volume of a cone Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Apply volume of solids <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners work in their groups, then as a class, to explain and apply the evidence. (1) Using 3 cones = 1 cylinder, they write $V(\text{cone}) = \frac{1}{3} \times \pi r^2 h$ and show it gives 616 cm ³ for the models. (2) They compare the one-third pattern with the pyramid and prism ratio from Lesson 5. (3) They calculate the drum volume: $V = \pi r^2 h = 22/7 \times 35 \times 35 \times 105 = 22/7 \times 1,225 \times 105 = 404,250 \text{ cm}^3 = 404.25 \text{ litres}$. (4) They recall the box volume of 322,500 cm ³ (322.5 litres) and compute the difference: 404,250 - 322,500 = 81,750 cm ³ = 81.75 litres more maize in the drum. (5) First confirming that both SAs equal 30,800 cm ² , they write a 3-4 sentence explanation for Mzee Kiplagat that explains why shape, not material alone, decides capacity, using the idea that a circular cross-section encloses more area for its perimeter than a square, and they distinguish surface area (material covering) from volume (space inside). (6) They answer an extension: a conical grain heap of $r = 1.4 \text{ m}$ and $h = 3 \text{ m}$ has what volume, and how many litres? ($22/7 \times 1.96 \times 3 \times \frac{1}{3} = 6.16 \text{ m}^3 = 6,160 \text{ litres}$).	Ask: 'We have 616 cm ³ and 1,848 cm ³ . What is the relationship, and what does it tell us about the cone formula?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners to complete the sentence 'The cone holds ____ of the cylinder, so $V = \frac{1}{3} \pi r^2 h$.' Then say: 'Mzee Kiplagat is waiting. Calculate the drum volume step by step. Show me $\pi r^2 h$ with numbers.' Before comparing volumes, insist: 'First, what are the surface areas? Say them aloud.' (30,800 cm ² each). Cold-call 2 learners to state the difference in litres (81.75). Ask the key question: 'If surface area is equal, why are the volumes not equal?' and wait 10-15 seconds, then collect 3 explanations and sharpen them: 'So surface area tells us how much sheet; volume tells us how much space inside.' Challenge: 'Would Wanjiru's customer prefer a sphere? What do you think?' Use only the established idea that a sphere is best of all for a fixed surface area, without deriving it. Correct any use of 'area' for 'volume' immediately.	Claim-Evidence-Reasoning (CER) with a number talk: the claim ('the drum holds more'), the evidence (404,250 cm ³ vs 322,500 cm ³ with equal 30,800 cm ²), and the reasoning (surface area and volume are different quantities; shape sets capacity) is written in full sentences, which forces learners to separate the two measures and close the loop on the phenomenon.	Collect each learner's CER paragraph as an exit check. Success indicators: (a) states both SAs as 30,800 cm ² ; (b) gives 404,250 cm ³ and 322,500 cm ³ ; (c) states 81.75 litres difference; (d) explicitly says surface area measures material and volume measures space inside; (e) uses the one-third rule correctly in the grain heap question (6.16 m ³). Tally learners who still say 'same surface area so same volume' and plan a short re-teach with a quick sand pour for them.
Driving Question Board (DQB) Creation  VIDEO: Volumes of cones intuition Source: Khan Academy (English - US curriculum)  Search ARES	Learners return to the Driving Question Board with the driving question: 'Why does the same amount of metal sheet make containers that hold very different amounts of grain — and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction?' In groups they place sticky notes in two	Say: 'Look at the board. Which questions can we now answer with evidence?' Ask each group to move one note and justify it. Cold-call 3 groups to read an answered question with its evidence. Give WAIT TIME of 10-15 seconds when asking: 'What new questions has today raised for you?' Invite 2 more learners to add new notes.	Driving Question Board with questions-to-evidence mapping: moving questions from 'unanswered' to 'answered' with cited data makes learning visible and lets learners own the next steps in the unit.	Count how many notes were moved with a cited piece of evidence (aim for at least one per group). Check that new questions use correct vocabulary (surface area vs volume, capacity, litres). Note learners who ask design-oriented questions as signals of readiness for Lesson 7 and 8.

for similar videos EXERCISE: Surface Area and Volume of Cylinders Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	columns: 'Answered today' (e.g., 'How much does the drum hold? 404,250 cm³ = 404.25 litres'; 'Why does the cone hold less than the cylinder? A third') and 'New questions' (e.g., 'Would a sphere be even better?'; 'How do we combine a cylinder and a cone, like the Rift Valley silo?'; 'How would Wanjiru choose dimensions for the least sheet?').	Point out that composite solids (Lesson 7) and design (Lesson 8) will address several of the new questions, without answering them now. Praise evidence-based wording, not just correct answers.		
Model Building Phase VIDEO: Volume of a sphere Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Surface Area and Volume of Cones Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners update the cumulative class model of the phenomenon on chart paper. The model has: (1) a labelled drawing of the drum ($r = 35$ cm, $h = 105$ cm) and box (50 cm x 50 cm x 129 cm); (2) the SA calculation 30,800 cm² for both; (3) new today: the volume of the drum 404,250 cm³ (404.25 litres) vs the box 322,500 cm³ (322.5 litres); (4) the volume formulas: cone $\frac{1}{3}\pi r^2 h$, sphere $\frac{4}{3}\pi r^3$, frustum big cone minus small cone; and (5) a sentence 'Equal surface area does not mean equal volume; shape decides capacity.' They add the matching models evidence (616 and 1,848 cm³) and the football and frustum examples as supporting evidence. They then revise their original predictions in their notebooks, writing 'My earlier idea ... My new idea ... Evidence ...'.	Say: 'Build the model so that someone from another school can see the answer to Mzee Kiplagat without us explaining.' Assign each group one section of the chart. Give WAIT TIME of 10-15 seconds before groups start so they can plan. Cold-call 3 learners to explain one part of the model to the class. Ask: 'What part of the driving question can we answer now, and what is still missing?' (Missing: combining solids and designing the best shape.) Ask 2 learners to read their revised ideas. Close with: 'Wanjiru measured, she did not guess. Tomorrow we combine solids like she does in her workshop.'	Cumulative Model Revision with Before-and-After Reflection: a model that links prediction, data, formula and conclusion lets learners see how their thinking changed, and it preserves one shared record to refine in Lessons 7 and 8.	Check that every group's model includes equal SA (30,800 cm²), both volumes, the difference of 81.75 litres and one volume formula with its evidence. Use the 'My earlier idea / new idea / evidence' entries to judge individual conceptual change. Success indicator: learners can say in their own words that surface area is the material and volume is the space inside.

D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners get 3 cone-fuls consistently, and did any groups' pour errors (not filled to the brim, mismatched r or h) create confusion that I used productively? (2) How many learners still treat surface area and volume as interchangeable, and which pour or sentence frame helped them? (3) Did learners connect the one-third rule to Lesson 5's pyramid ratio, or did they see it as a new, separate fact? (4) Could groups explain why the drum holds 81.75 litres more using the circular cross-section idea, or did they only quote numbers? (5) Which questions on the DQB signal readiness for composite solids in Lesson 7? Adjust the grouping and the language support for Lesson 7 accordingly.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that 3 cone-fuls of water or sand fill a cylinder with the same radius (7 cm) and height (12 cm), so the cone holds 616 cm ³ and the cylinder 1,848 cm ³ . We also saw the football (r = 10.5 cm) hold 4,851 cm ³ and the Turkana frustum tank hold 11.319 m ³ . The drum holds 404,250 cm ³ and the box 322,500 cm ³ , although both used 30,800 cm ² of sheet.
What did I learn?	We learned that $V(\text{cone}) = \frac{1}{3}\pi r^2 h$, $V(\text{sphere}) = \frac{4}{3}\pi r^3$, and $V(\text{frustum}) = \text{big cone minus small cone}$. We learned that surface area measures the material covering a solid, while volume measures the space inside, so they are different quantities.
How does this explain the phenomenon?	We explained that the drum holds 81.75 litres more maize than the box because, with the same surface area of 30,800 cm ² , the drum's shape (circular cross-section) encloses more space than the box's square cross-section. Shape, not only the amount of material, determines capacity, so Wanjiru can use mathematics to choose shapes that give more capacity for the same sheet.

LESSON 7 (80 minutes (2 x 40-minute periods)): Composite Solids: Volume, Surface Area & Real-Life Applications

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners combine surface area (the material or paint needed to cover a solid) and volume (the capacity inside it) in one design context. They decide which quantity a question needs and use this to advance the driving question about how a Kenyan artisan designs the best solid for storage or construction.
Knowledge	By the end of the lesson the learner should be able to: (a) determine the surface area of composite solids (cylinder with hemispherical top, cylinder with cone roof); (b) determine the volume of composite solids; (c) decide whether a real-life question needs surface area (material, painting cost) or volume (capacity in litres); (d) convert m ³ to litres (1 m ³ = 1,000 litres); (e) appreciate the use of surface area and volume of solids in real-life situations. Key inquiry questions: How do we determine the surface area and volume of solids? How are surface area and volume of solids applied in real-life situations?
Skills	Analysing and interpreting (decomposing a composite solid into cylinder, hemisphere, cone); using mathematics and computational thinking (multi-step calculation with $\pi = 22/7$); constructing explanations (justifying whether a question needs SA or volume); developing and using models (revising the class model of the drum and box).
Attitudes	Precision and integrity in recording calculations; appreciation of how mathematics supports Kenyan water and grain projects; collaboration and respect for peers' reasoning (Core competencies: Critical thinking and Problem Solving; Communication and Collaboration. Values: Responsibility, Integrity. PCIs: Education for Sustainable Development; Life skills (decision making)).
Key Inquiry Question	How do we determine the surface area and volume of solids, and how are they applied in real-life situations such as water tanks and grain silos in Kenya?
Purpose in Storyline	Lessons 1-6 gave the formulae. Lesson 4 covered the SA of spheres and composite solids, and Lesson 6 covered the volumes of cones, frustums and spheres, which resolved why the drum holds more than the box at equal surface area. Lesson 7 uses both quantities together in the same design (a hemispherical-top tank and a Rift Valley silo). Learners must choose between SA (sheet, paint, cost) and volume (litres, grain) for each question. This prepares them to optimise and present their own container designs in Lesson 8.
Safety Notes	Keep walkways clear when learners walk around the measuring stations. Use dry sand or rice only for model-filling and sweep spills at once. Handle metal tins and cardboard models carefully because edges may be sharp. No learner climbs on desks or real tanks during the field-link discussion.

B. LESSON OVERVIEW

Mzee Kiplagat's posho mill in Eldoret has shown that the drum and the box, each made from 30,800 cm² of iron sheet, hold different amounts of maize (404,250 cm³ against 322,500 cm³). Now the Rift Valley cooperative asks for two real jobs: a water tank and a grain silo. Learners first predict whether painting cost and storage capacity depend on the same calculation. They then work through a hemispherical-top water tank (cylinder $r = 2.1$ m, $h = 4$ m, flat base) to find $SA = 94.38$ m² and $V = 74.844$ m³ = 74,844 litres. Next they model a Rift Valley silo (cylinder $r = 2.1$ m, $h = 6$ m, cone roof of height 2.8 m and slant 3.5 m) to find $TSA = 116.16$ m² and $V = 96.096$ m³. In the Explain phase learners sort questions into SA-questions and volume-questions, and justify why they are not interchangeable. They then add new entries to the Driving Question Board (DQB) and revise the class model in preparation for Lesson 8. All figures use $\pi = 22/7$.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Composite Solids Principles - Basic Source: CK-12  Search ARES for similar videos  EXERCISE: Volume and surface area word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the posho mill puzzle: the drum and the box each used 30,800 cm² of sheet, yet the drum holds 404.25 litres and the box 322.5 litres. The teacher shows a photo of a cylindrical water tank with a dome top at a Rift Valley cooperative and tells the story: 'The cooperative must paint the tank and also tell farmers how many litres it holds.' Individually, learners write a prediction in their books: (1) Will the paint needed and the water held come from the same calculation? (2) Which part of the tank will need more sheet, the wall or the dome? Then they sketch the tank and label the parts they think they will need to calculate. Pairs share, then the class records a few ideas on a chart titled 'Our predictions about the tank'.	Display the tank photo and say: 'Mzee Kiplagat's drum and box used equal sheet but held different grain. Today a cooperative in Nakuru asks us for TWO answers about one tank: how much paint, and how much water. Write your prediction first, silently.' Give WAIT TIME of 10-15 seconds after each question before any learner speaks. Then ask: 'Will paint and water come from the same calculation? Why?' Cold-call 4 learners (including at least 2 who have not volunteered) and record their reasoning verbatim on the chart without correcting. Ask 2 more learners: 'Which would you expect to be larger in number, the sheet in m² or the capacity in m³? Can you even compare them?' Circulate and read predictions, noting who writes 'SA and volume are the same thing', and tell the class: 'We will test every prediction with evidence today.'	Predict-Observe-Explain (POE) with a written, committed prediction. Writing a prediction before calculating exposes learners' conception that surface area and volume are interchangeable, and gives them something concrete to confirm or revise when the evidence arrives.	Look for: (a) learners who write a prediction with a reason instead of a blank or a one-word answer; (b) learners who use the words 'surface area' and 'volume' correctly or who confuse them; (c) the count of learners (record a tally) who predict that paint and water come from the same calculation. Use the tally later to compare with the evidence.
Observe Phase  VIDEO: Composite Solids Principles - Basic Source: CK-12  Search ARES for similar videos  EXERCISE: Volume and surface area word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners work through the Evidence Card: Tank and Silo. Station A (Water tank): a cylinder of $r = 2.1$ m and $h = 4$ m with a hemisphere on top and a flat base. Using a tin-and-ball model (or a cardboard net) they decompose the tank into the cylinder wall, the base and the hemispherical dome. They calculate: cylinder wall = $2 \times 22/7 \times 2.1 \times 4 = 52.8$ m²; base = $22/7 \times 2.1 \times 2.1 = 13.86$ m²; hemisphere curved SA = $2 \times 22/7 \times 2.1 \times 2.1 = 27.72$ m². Total SA = 94.38 m². They then calculate the volume: cylinder = $22/7 \times 2.1 \times 2.1 \times 4 =$	Say: 'Draw the solid first, then split it into pieces you already know. Which faces are on the outside? Which are hidden?' Circulate to Station A and ask: 'Why is there no circle between the cylinder and the dome in your surface area?' Give WAIT TIME of 10-15 seconds, then cold-call 3 groups for their answer. Ask at Station B: 'The roof uses slant height 3.5 m. Where does the 2.8 m go?' (Check: 2.1 squared plus 2.8 squared equals 3.5 squared.) Cold-call 2 learners to say which height goes into the cone volume and which length goes into the cone's curved area.	Decomposition and a Two-Column Evidence Table (SA vs Volume). Splitting a composite solid into known solids shows which faces count for covering and which solids count for filling. A table with units forces learners to see that the same solid yields two different quantities.	Observable indicators: (a) learners list exactly 3 surface pieces for the tank and 3 for the silo, excluding the hidden joining circles; (b) they use slant height 3.5 m for the roof area and vertical height 2.8 m for the cone volume; (c) correct totals: tank 94.38 m² and 74.844 m³, silo 116.16 m² and 96.096 m³; (d) correct units written on every line. Use a quick checklist per group; give a hint card to groups with fewer than 3 of 4 indicators.

	55.44 m³; hemisphere = $\frac{2}{3} \times \frac{22}{7} \times 2.1^3 = 19.404$ m³; V = 74.844 m³ = 74,844 litres. They record that there is no flat circle between the cylinder and dome because that circle is inside the tank and is not covered with sheet. Station B (Rift Valley silo): cylinder r = 2.1 m, h = 6 m, cone roof of height 2.8 m and slant 3.5 m, flat floor. Learners calculate the wall = $\frac{22}{7} \times 2.1 \times 3.5 = 23.1$ m², floor = 13.86 m²; total SA = 116.16 m². Volume = cylinder 83.16 m³ + cone 12.936 m³ = 96.096 m³. They record every step in a two-column table (Surface Area of tank / silo; Volume of tank / silo) with units. Learners fill model silos with rice to see capacity and wrap them with paper to see covering material.	Remind groups to keep pi = $\frac{22}{7}$ and to round only at the last step. Pause the class at the halfway point: 'Hold up fingers: how many parts does your tank SA have? (3). How many parts does your silo SA have? (3).' Collect any miscounts and re-teach quickly with the model. Ask about the units: 'Is 94.38 a number of m² or m³? How do you know?'		
Explain Phase  VIDEO: Cylinder volume & surface area <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Volume and surface area of cylinders <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners use their tables to answer a Question Sort. They receive 8 strips, e.g.: 'How many litres of water does the tank hold?'; 'How much paint is needed for the tank?'; 'How much sheet is needed for the silo roof?'; 'How many bags of maize can the silo store?'; 'What is the cost of painting the outside at KES 150 per m²?'; 'How much grain fits in the silo?'. Groups sort each strip into SURFACE AREA or VOLUME, solve at least 3 of them using their table, and write a one-line reason for each. Worked tasks: (i) Water: 74.844 m³ x 1,000 = 74,844 litres; (ii) Painting cost of the tank = 94.38 x 150 = KES 14,157 for the outer cover (state the assumption clearly that the whole 94.38 m² is painted); (iii) Silo grain capacity = 96.096 m³ and sheet for the whole silo = 116.16 m². Learners then return to the posho mill: they explain, in 3	Say: 'Read the question and underline the word that tells you what it is about: paint, sheet, cover = surface area; litres, capacity, store, fill = volume.' After groups finish, give WAIT TIME of 10-15 seconds and ask: 'Which strip was hardest to sort, and why?' Cold-call 4 groups to defend their sorting and invite 2 other groups to challenge: 'Do you agree? What evidence from your table supports that?' Press for the key idea: 'A bigger surface area does not tell us a bigger volume, and the drum and box prove it.' Ask: 'The silo needs 116.16 m² of sheet and holds 96.096 m³. The tank needs 94.38 m² and holds 74.844 m³. Which uses less sheet for each cubic metre stored? Show it.' (silo: $116.16 / 96.096 = \text{about } 1.21$ m² per m³; tank: $94.38 / 74.844 = \text{about } 1.26$ m² per m³). Note that dividing is an extension, so let	Claim-Evidence-Reasoning (CER) with a Question Sort. Learners make a claim (SA or volume), cite numbers from their tables as evidence, and give reasoning linked to the definitions. Sorting makes explicit the idea that surface area and volume answer different questions even for the same solid.	Look for: (a) at least 7 of 8 strips sorted correctly; (b) reasoning that names what is measured (covering material vs space inside); (c) conversion of 74.844 m³ to 74,844 litres without error; (d) a statement that equal surface area does not mean equal volume, citing the drum and box numbers. Record who needs support with m³ to litres and revisit it at the start of the next period.

	sentences, why Mzee Kiplagat cannot predict capacity from sheet area alone, using the example that the drum and box have equal SA (30,800 cm²) but volumes of 404,250 cm³ and 322,500 cm³. They also compare their original predictions against the evidence.	strong groups explore it. Invite 3 learners to revisit the Predict chart and mark each prediction 'supported' or 'revised', with one piece of evidence for each.		
Driving Question Board (DQB) Creation  VIDEO: Surface area of a box (cuboid) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Apply volume of solids <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the class Driving Question Board with the driving question displayed: 'Why does the same amount of metal sheet make containers that hold very different amounts of grain, and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction?' Each pair writes one sticky note: either an answer or a new question arising from today, e.g. 'Which shape holds the most for a fixed sheet area?', 'Why does the silo have a cone on top?', 'How can we cut the cost of painting?'. They stick answers to the existing questions and move earlier questions into 'Answered', 'Partly answered' or 'New questions' columns. As a class they discuss which questions will be used in Lesson 8.	Say: 'Look at the board. Which of our questions can we now answer with evidence from today? Walk up and move one.' Give WAIT TIME of 10-15 seconds, then cold-call 5 pairs to read their note aloud and explain which number or calculation supports it. Ask: 'What does a Kenyan metal fabricator like Wanjiru need to know before she cuts the sheet?' Elicit: capacity needed (volume), sheet available (surface area), and cost. Highlight 2 unanswered questions (e.g. the best shape for a fixed sheet) and tell the class these will shape their Lesson 8 designs. Praise notes that cite evidence, and ask the authors of vague notes to add a number or a unit.	Driving Question Board with Question Sorting. Public tracking of answered and unanswered questions makes progress visible, gives learners ownership of the inquiry, and links today's tasks directly to the driving question.	Observable: each pair posts at least 1 note; at least 3 notes refer to a specific figure (such as 94.38 m², 74,844 litres or 96.096 m³); learners move questions to the correct column and justify the move aloud. Note learners who only restate facts and prompt them to write a 'why' or 'how' question.
Model Building Phase  VIDEO: Metric system: units of volume <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners revise the class model of 'Surface area vs volume' (poster or concept map). Pairs add a new section called 'Composite solids': a labelled sketch of the tank and silo, the pieces used for SA (wall, base, dome or roof) and the pieces used for volume (cylinder plus hemisphere or cone), and a decision rule: 'Material, paint, cost = SA (m²). Capacity, grain, water = volume (m³, litres).' They connect this to the central model of	Say: 'Add to our model: what new pieces did composite solids give us?' Give WAIT TIME of 10-15 seconds, then cold-call 4 learners to add one element each to the poster and invite 2 others to correct or extend. Ask: 'Where on our model do we show that surface area and volume are not interchangeable?' Check that learners link the decision rule to the posho mill numbers. Prompt: 'If the cooperative has only 100 m²	Cumulative Model Revision (concept map). Adding composite solids and a decision rule to a shared model consolidates formulae from Lessons 4 and 6 and prepares learners for the design and critique work in Lesson 8.	Check exit statements for: (a) correct pairing of 'surface area' with material or cost and 'volume' with capacity; (b) reference to the drum and box or the tank and silo as evidence; (c) a project idea with a named community need. Sort statements into 'secure', 'nearly there' and 'needs reteaching' to plan the start of Lesson 8.

 EXERCISE: Surface Area and Volume of Cones Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	the drum and the box: equal SA 30,800 cm² but volumes 404,250 cm³ and 322,500 cm³. Each learner then writes an 'exit statement' completing the sentence: 'To design a good container, an artisan must calculate both ____ and ____ because ____.' They also draft one idea (shape, dimensions and need) for their Lesson 8 project, such as a grain silo, water tank or sufuria for a named community.	of sheet, would you check surface area first or volume first? Why?' (Always compare SA before comparing volumes.) Collect exit statements. Tell groups to bring dimensions and a community need for Lesson 8.		
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners correctly leave out the hidden circle where the hemisphere or cone joins the cylinder? Which learners still included it, and what model or sketch helped them most? (2) Did they choose slant height for curved area and vertical height for volume of the cone roof without prompting? (3) Could they explain, with the drum and box numbers, that equal surface area does not give equal volume? (4) How many errors came from converting m³ to litres (1 m³ = 1,000 litres) or from unit slips between m² and m³? (5) Which DQB questions need to be carried into Lesson 8? Adjust grouping and prepare short reteaching cards for the learners who struggled.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the hemispherical-top water tank (cylinder $r = 2.1$ m, $h = 4$ m) has a total surface area of 94.38 m ² and a volume of 74.844 m ³ (74,844 litres). We also observed that the Rift Valley silo (cylinder $r = 2.1$ m, $h = 6$ m with a cone roof) has a total surface area of 116.16 m ² and a volume of 96.096 m ³ .
What did I learn?	We learned that a composite solid is split into known solids; for surface area we add only the outside faces (the joining circles are hidden), and for volume we add the volumes of the parts. Surface area (m ²) tells us the material or paint needed, and volume (m ³ or litres) tells us the capacity, so we choose the one that the question needs. 1 m ³ = 1,000 litres.
How does this explain the phenomenon?	We explain that the drum and the box in Mzee Kiplagat's posho mill have the same surface area (30,800 cm ²) but different volumes (404,250 cm ³ and 322,500 cm ³) because shape, not just the amount of sheet, decides capacity. A Kenyan artisan therefore calculates both surface area and volume, as for the tank and silo, to choose a shape and dimensions that give the capacity needed for the least sheet and cost.

LESSON 8 (80 minutes (2 x 40-minute lessons)): Project Presentations, Model Building & Unit Consolidation: Why the Drum Holds More Than the Box

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners communicate and critique an optimised container design (dimensions, material, capacity) and co-construct the final DQB model that fully answers the driving question about Mzee Kiplagat's drum and box.
Knowledge	By the end of the sub-strand the learner should be able to: determine the surface area of prisms, pyramids, cones, frustums and spheres, and composite solids; determine the volume of prisms, pyramids, cones, frustums, spheres and composite solids; apply surface area and volume of solids to real-life situations; appreciate the use of surface area and volume of solids in real-life situations. In this lesson learners consolidate Lessons 1-7: the drum and the box both have TSA 30,800 cm ² , but volumes of 404,250 cm ³ (404.25 litres) and 322,500 cm ³ (322.5 litres). Surface area is the material covering a solid; volume is the space inside. For a fixed surface area, shape determines capacity.
Skills	Communication and Collaboration: present a design with dimensions, material (TSA) and capacity, and give and receive critique. Critical Thinking and Problem Solving: check a design's calculations and decide whether a question needs surface area or volume. NGSS Practices: Constructing Explanations, Engaging in Argument from Evidence, Developing and Using Models, Communicating Information.
Attitudes	Integrity: reporting measurements and calculations honestly and accurately. Responsibility: caring for models and resources. Unity: respecting every group member's input. Appreciating how mathematics helps Kenyan artisans such as Wanjiru in Eldoret.
Key Inquiry Question	1. How do we determine the surface area and volume of solids? 2. How are surface area and volume of solids applied in real-life situations?
Purpose in Storyline	This is the final lesson. Lessons 1-7 gave the pieces: the sum-of-faces and net method for prisms, the pyramid, cone and frustum areas, the sphere and composite SA, the volume of prisms, pyramids, cones, frustums and spheres (including the drum's 404,250 cm ³ against the box's 322,500 cm ³), and composite modelling. Here learners use these to present their own container designs and complete the DQB with a final explanation of why equal sheet gives unequal capacity.
Safety Notes	Keep walkways clear during gallery presentations. Carry models carefully. Use only blunt tools for model cutting. Do not stand on desks or chairs to hang posters. Any pouring of grain or sand is done on a tray, and spills are cleaned immediately to prevent slips.

B. LESSON OVERVIEW

Mzee Kiplagat's posho mill in Eldoret has a drum and a box, both closed and made by Wanjiru from exactly 30,800 cm² of iron sheet. Yet the drum ($r = 35$ cm, $h = 105$ cm) holds 404,250 cm³ = 404.25 litres, while the box (50 cm x 50 cm x 129 cm) holds 322,500 cm³ = 322.5 litres, so the drum holds 81.75 litres more maize. In this final lesson, groups present their project, a grain silo, water tank or sufuria designed for a named Kenyan community need, with dimensions, TSA and capacity. Peers critique using evidence. The class then builds the final DQB model, compares it with the Lesson 1 model, writes the Final Explanation, and links the mathematics to careers such as metal fabricator, civil engineer, architect, packaging engineer and quantity surveyor.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Cylinder volume & surface area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Apply volume of solids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look at the Lesson 1 initial model, in which many assumed equal sheet means equal grain. Individually, in 3 minutes, they write a prediction: 'Which of the group designs will hold the most for its sheet metal, and what feature of the shape will explain it?' They then pair-share and note the prediction on a sticky note. They also predict: if Wanjiru made a sphere-shaped tank from 30,800 cm ² , would it hold more or less than the drum? They record a reason in one sentence.	Display the Lesson 1 model and the photographs of the drum and the box. Say: 'Remember Mzee Kiplagat's puzzle. Before we present, I want your prediction, not an answer.' Give 3 minutes of silent writing, then WAIT TIME of 10-15 seconds before calling on anyone. Cold-call 4 learners: 'Tell us your prediction and the reason behind it.' Record 4 predictions on the board without judging. Ask: 'Is anyone still thinking that equal sheet must mean equal grain? What would change your mind?' Say: 'Keep your sticky note; we will return to it at the end.'	Predict-Observe-Explain (POE) with revisiting of the initial model: committing to a prediction first makes learners aware of their prior ideas, so later evidence from the presentations creates real conceptual change instead of passive acceptance.	Observe whether learners (a) name shape as a factor, not only material, (b) use both terms 'surface area' and 'volume' correctly and separately, and (c) give a reason with a feature such as a circular cross-section. Tally on a checklist how many still treat equal surface area as equal volume. Target: at least 20 of 40 learners give a reasoned prediction that mentions shape.
Observe Phase  VIDEO: Cylinder volume & surface area Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Volume of cylinders, spheres, and cones word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups give 4-minute project presentations (grain silo for a cooperative in Nakuru or Uasin Gishu, water tank for a Turkana or Kajiado community, or a sufuria for a school kitchen), followed by 2 minutes of critique. Each presentation states: (1) the community need, (2) dimensions, (3) the TSA of material, saying whether it is total or curved, (4) the volume converted to litres (1 m ³ = 1,000 litres; 1 litre = 1,000 cm ³), (5) one reason for the chosen shape. Presenters use models, posters and nets. The Rift Valley silo (116.16 m ² , 96.096 m ³) and the hemispherical-top tank (94.38 m ² , 74.844 m ³) from Lesson 7 serve as benchmark designs. Observers fill a Critique Sheet recording each group's TSA, capacity, and a litres-per-m ² of sheet figure. They check one calculation from each design independently and give 'two stars and a wish'.	Set the rules: 'Critique the mathematics and the design, not the person.' Run a timer. After each presentation, WAIT TIME of 10-15 seconds before taking questions. Prompt: 'Which formula did you use for the curved part, and did you include the lid or floor?' 'Did you use slant height l or vertical height h for the cone roof?' Cold-call 3 observers per presentation: 'Give one star and one wish, with the figure you checked.' Circulate and verify the arithmetic with $\pi = 22/7$. Where a group confuses surface area with volume, ask: 'Is this the material or the space inside?' Ensure that units are cm ² , m ² , cm ³ and m ³ and that rounding is done only at the final step to 2 decimal places.	Gallery-style presentation with Evidence-Based Peer Critique (Engaging in Argument from Evidence): learners must support or challenge a design with calculations, which strengthens their command of formulae and transfers the drum-vs-box reasoning to new contexts.	Use a 4-point checklist per group: dimensions stated, TSA correct and labelled total or curved, volume correct and converted to litres, shape justified. Record observable indicators such as 'uses l for the curved area and h for volume', 'checks the other group's arithmetic', 'distinguishes material from capacity'. Flag groups that swap SA and V terms for immediate feedback. Expect at least 70% of groups to hit 3 of 4 indicators.

Explain Phase  VIDEO: Surface area of a box using nets <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Apply volume of solids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the anchor data and write the Final Explanation in groups of 4, using the Claim-Evidence-Reasoning (CER) frame. Claim: the drum holds more than the box even though both use the same sheet. Evidence: both closed containers have TSA 30,800 cm² (drum: $2 \times 22/7 \times 35 \times 35 + 2 \times 22/7 \times 35 \times 105 = 7,700 + 23,100$; box: $2(50 \times 50) + 4(50 \times 129) = 5,000 + 25,800$); the drum's volume is $22/7 \times 1,225 \times 105 = 404,250$ cm³ = 404.25 litres and the box's is $50 \times 50 \times 129 = 322,500$ cm³ = 322.5 litres, a difference of 81,750 cm³ = 81.75 litres. Reasoning: surface area measures the material covering a solid and volume measures the space inside; they are different quantities. For a fixed surface area, the circular cross-section encloses more area for its perimeter than a square, so the cylinder holds more. A sphere is best of all, so artisans choose cylinders, domes and cones for a large capacity with little sheet metal, balanced against stacking and cost. Learners then apply this to their own design: could a change of shape increase capacity for the same sheet? They also state which of the two measures (SA or V) answers each of these questions: how much paint? how many sacks of maize? how much sheet for Wanjiru to buy?	Say: 'Now write the explanation Mzee Kiplagat can understand. Use numbers, not just words.' Give 6 minutes for writing, then WAIT TIME of 10-15 seconds. Ask: 'Why compare surface areas before comparing volumes?' and 'Which part of your reasoning says shape, not material, decides capacity?' Cold-call 5 groups to read one sentence each and have the class build the explanation sentence by sentence. Probe misconceptions: 'Two learners said bigger surface area means bigger volume; use the drum and box to test that.' Ask: 'A sphere is best of all, so why does Wanjiru not make spherical tanks for grain?' Guide them to practical needs such as stacking, flat floors, cost and welding. Check that learners do not re-derive formulae but use them as tools.	Claim-Evidence-Reasoning (CER) with Constructing Explanations: requiring the claim to be supported by the calculated figures and a general principle pushes learners from describing a result to explaining the mechanism (shape determines capacity for fixed material), and integrates all lessons.	Score each written explanation with a 3-level rubric: (1) states difference only; (2) cites the 30,800 cm² and the two volumes; (3) also explains that SA and V are different quantities and that shape (circular cross-section) drives capacity. Observable indicators: correct units, correct litres conversion, no interchange of 'surface area' and 'volume', mentions the sphere as the best shape. Target: at least 75% of groups at level 2 or 3, and all learners can say which measure to use for paint or sheet versus grain.
Driving Question Board (DQB) Creation  VIDEO: Surface area of a box (cuboid) <i>Related topic: not</i>	The class revisits the DQB built since Lesson 1. Groups take the sticky-note questions (for example 'Does equal sheet give equal capacity?', 'Why does the drum hold more?', 'How do we calculate the sheet for a cone roof?') and move each one to a column:	Say: 'Our board is the record of what we wondered and what we now know.' Read each question aloud. After each, WAIT TIME of 10-15 seconds, then ask: 'What evidence answers this?' Cold-call 6 learners to place and justify questions. Push on partly	Driving Question Board sorting (Asking Questions and Defining Problems): revisiting and classifying questions makes learning progress visible, shows what evidence resolved each question, and keeps open questions for further inquiry, linking	Check that learners cite specific evidence for answered questions (for example 'Lesson 6: drum 404,250 cm³ vs box 322,500 cm³'). Observable indicators: accurate placement of questions, at least 2 new meaningful questions per group, a career

<i>an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Volume of cylinders, spheres, and cones word problems Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Answered, Partly Answered, New Question. For each Answered question, a learner attaches the evidence (lesson, formula or figure). They write 2 new questions arising from the project, such as 'Would a cylinder with a different r and h hold even more for 30,800 cm²?' or 'How do costs of welding change the best design?'. They also add a career card linking the mathematics to a metal fabricator, civil engineer, architect, packaging engineer or quantity surveyor.	answered ones: 'What is still missing?' Ask: 'Which career would use this mathematics tomorrow morning?' Invite a learner who has a relative in jua kali, building or manufacturing to share. Photograph the board for the class portfolio.	the unit back to the driving question.	linked to the correct measure (SA for sheet or paint, V for capacity). Note learners who cannot attach evidence for follow-up.
Model Building Phase VIDEO: Surface area using a net: triangular prism <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Volume and surface area of cylinders <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Each group revises its Lesson 1 initial model into a final cumulative model on poster paper. The model has four linked parts: (1) the drum and the box drawn with dimensions and nets; (2) SA = material (30,800 cm² for both) and V = space inside (404,250 vs 322,500 cm³), clearly separated; (3) a formula toolkit from Lessons 1-7: prism TSA = 2 x base area + perimeter x length; pyramid TSA; cone curved SA = πrl and TSA = $\pi r(r + l)$; frustum curved SA = $\pi(R + r)l$; sphere SA = $4\pi r^2$ and V = $\frac{4}{3}\pi r^3$; prism/cylinder V = base area x h; pyramid/cone V = $\frac{1}{3}$ x base area x h; frustum V = big cone minus small cone; (4) the principle: for a fixed surface area, shape determines capacity, with the circular cross-section beating the square and the sphere best of all, balanced against stacking and cost. Groups place the Lesson 1 model beside the final model and write three 'what changed in my thinking' statements. They finish by completing the Summary Table	Say: 'Put your Lesson 1 model next to your final one. What did you believe then, and what do you know now?' Give 12 minutes for building, circulating with the prompts: 'Where does your model show the difference between material and space?' 'Where is the 81.75 litres?' 'Where does shape enter?' WAIT TIME of 10-15 seconds after each question. Cold-call 4 groups to share their change statements. Do a class vote: 'Show with fingers: 1 = still unsure to 4 = I can explain it to Mzee Kiplagat.' Close by answering the driving question in one voice: 'Equal sheet does not mean equal grain: surface area is the material and volume is the space, and the shape decides how much space that material can enclose. Artisans use mathematics to choose the shape and dimensions.' Collect the Summary Tables.	Cumulative Model Revision with Before-and-After Comparison (Developing and Using Models): placing the first model beside the final one makes the conceptual shift explicit, and requiring the model to include both quantities, the formula toolkit and the shape principle consolidates the unit into one coherent explanation of the phenomenon.	Use a model rubric: includes both containers with correct dimensions; separates SA (30,800 cm²) from V (404,250 vs 322,500 cm³); includes at least five correct formulae with l vs h distinguished; states the shape principle; shows 3 specific changes from Lesson 1. Observable indicators: correct units, correct litres conversions, the sphere mentioned as best, practical trade-offs mentioned. Expect at least 80% of models to meet 4 of 5 criteria. Collect exit tickets: 'Which measure would Wanjiru use to order sheet metal, and which would Mzee Kiplagat use to decide how many sacks fit?'

	(observed, learned, explained) and revisiting their opening sticky-note prediction.			
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D. TEACHER REFLECTION
Which groups still blurred surface area and volume, and which feedback or question helped them? Did learners use the drum-vs-box evidence (30,800 cm ² equal; 404,250 vs 322,500 cm ³) to justify their own design choices? How well did the peer critique stay on the mathematics? What would I change about time allocation for presentations versus the Final Explanation? Which learners showed growth between the Lesson 1 model and the final model, and what evidence supports this? Which new DQB questions could seed a future optimisation or cost-based investigation?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What we observed: Two closed containers made from the same 30,800 cm ² of iron sheet, a drum (r 35 cm, h 105 cm) and a box (50 cm x 50 cm x 129 cm), held different amounts of maize, and the groups' own container designs showed the same pattern: the shape changed the capacity.
What did I learn?	What we learned: Surface area measures the material covering a solid and volume measures the space inside; the drum holds 404,250 cm ³ (404.25 litres) and the box 322,500 cm ³ (322.5 litres), so the drum holds 81.75 litres more; we can find both for prisms, pyramids, cones, frustums, spheres and composite solids.
How does this explain the phenomenon?	What we can explain: For a fixed surface area, shape decides capacity because a circle encloses more area for its perimeter than a square, and a sphere is best of all, so artisans like Wanjiru use mathematics to choose shapes and dimensions that give the largest capacity for the least sheet metal, balanced against stacking and cost.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.